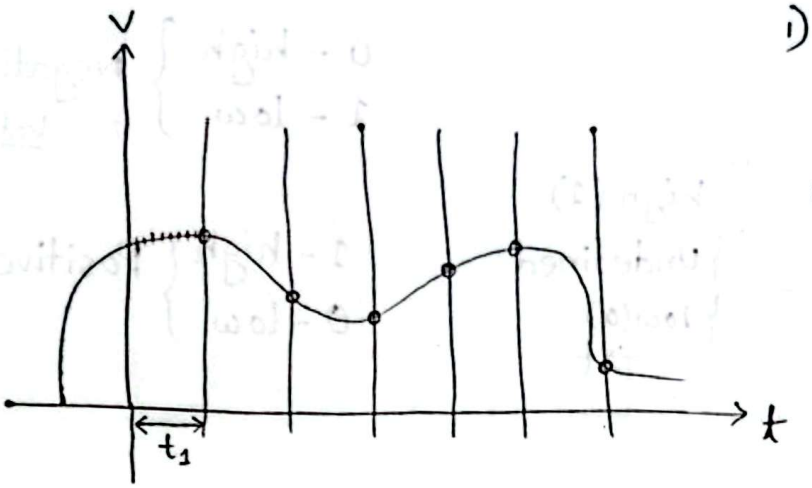


i) Digital Fundamentals
-Thomas L. Floyd

ii) Digital Systems
-Ronald J. Tocci

Analog → Continuous value

Digital → Discrete (transistor → discrete device)

Repeated value after a certain interval → discrete

Satallite tracking, space → Analog computer

Speaker → Analog signal (Converted to digital)

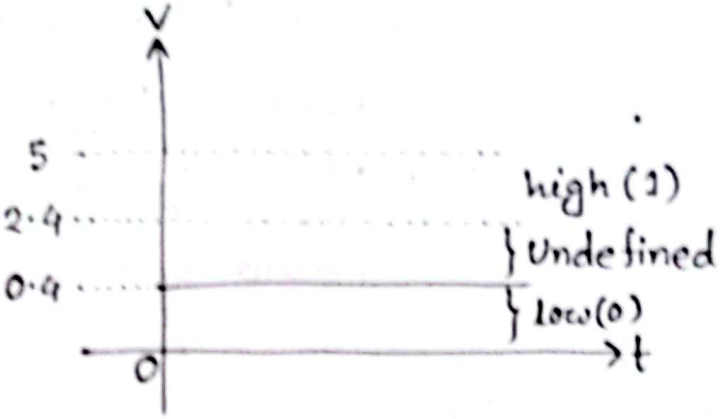
LSI → Large Scale Integration

VLSI → Very Large Scale Integration (hundred thousand components in a single IC)

ULSI → Ultra Large Scale Integration (million components in a single IC.)

- Binary Digit : Binary digits are 0 and 1 which are the digits in Binary System.
- Logic Levels :
- Digital Waveforms
- Timing Diagram

CamScanner

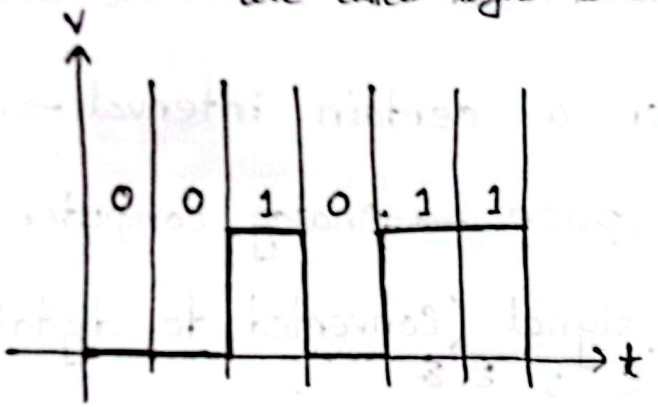


0 - high } Negative Logic
1 - low } Voltage

1 - high } Positive logic
0 - low }

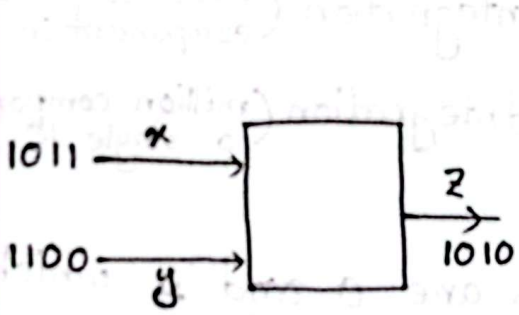
Logic Level

The voltages used to represent a 1 and a 0 are called logic levels



Digital waveforms carries binary data

Digital waveforms



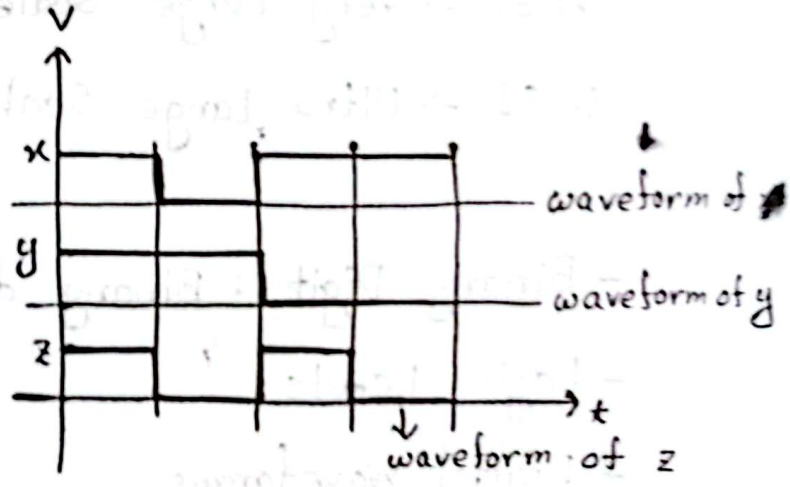
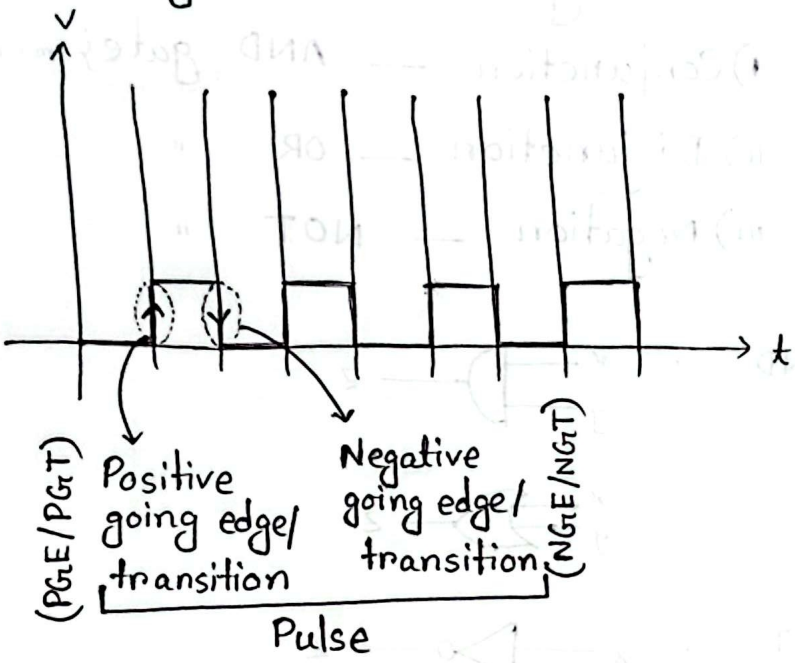


Fig : Timing diagram of D

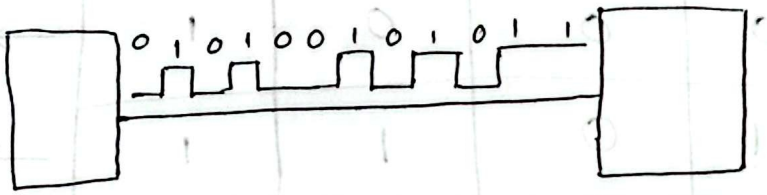
Clock Signal :



Lab : [Expression + mimize + truth table + ckt]

1. Verification of different logic gates using IC

serial data transfer



12 clock pluse

Parallel data transfer

3 clock pulse

Basic Logic Function :

i) Conjunction — AND (gate) → mainly transistor

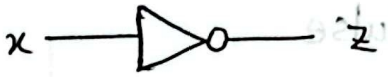
ii) Disjunction — OR "

iii) Negation — NOT "

AND :

OR :

NOT :



Truth table

X	Y	AND	OR	NOT
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

Boolean expression

→ $z = xy$ (and)

→ $z = x + y$ (or)

→ $z = \bar{x}$ (not)

Combinational Logic Gate

NAND : or

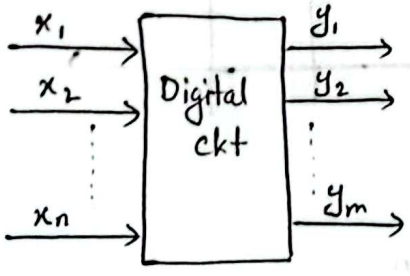
NOR : or, $z = \overline{x+y}$

XOR : $z = x \oplus y = x\bar{y} + \bar{x}y$

truth table

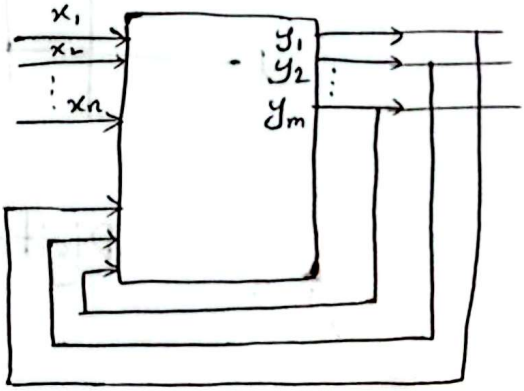
x	y	NAND	NOR	XOR
0	0	1	1	0
0	1	1	0	1
1	0	1	0	1
1	1	0	0	0

Digital circuits
1. Combinational Circuits
2. Sequential Circuits
} differences



Combinational ckt

(No storage Capacity)



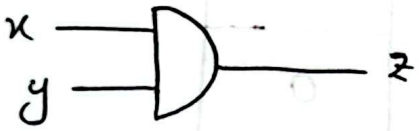
Sequential ckt

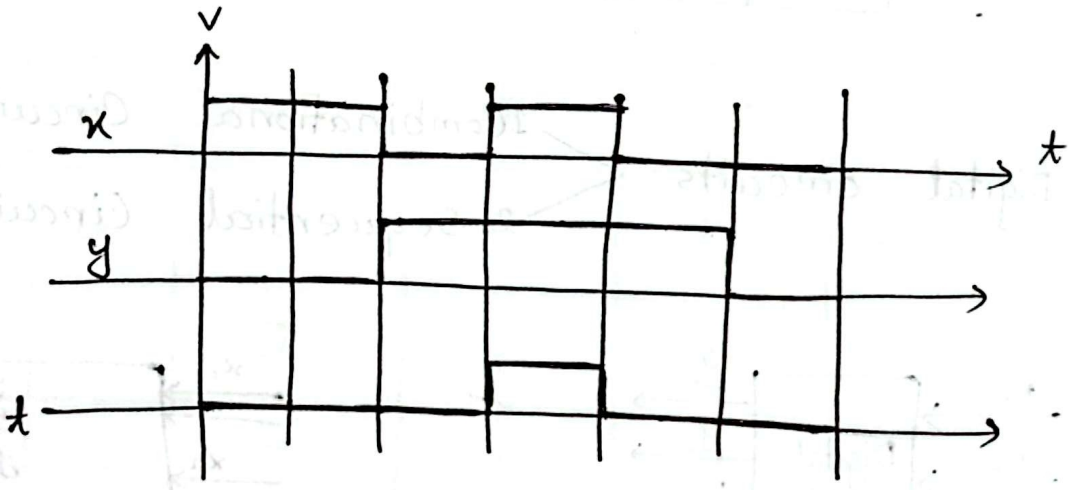
(Storage capacity)

Logic Functions

Combinational
- i) Addition + Substraction (Arithmatic)
  multiplication+division
- ii) Comparison
- iii) Code conversion
- iv) Encoding and Decoding
- v) Data Selection (Multiplexing + Demultiplexing)

Sequential
- vi) Storage Functions

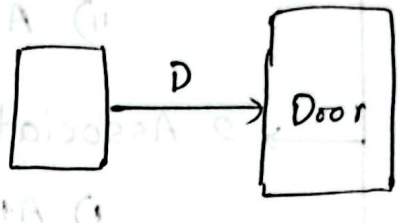




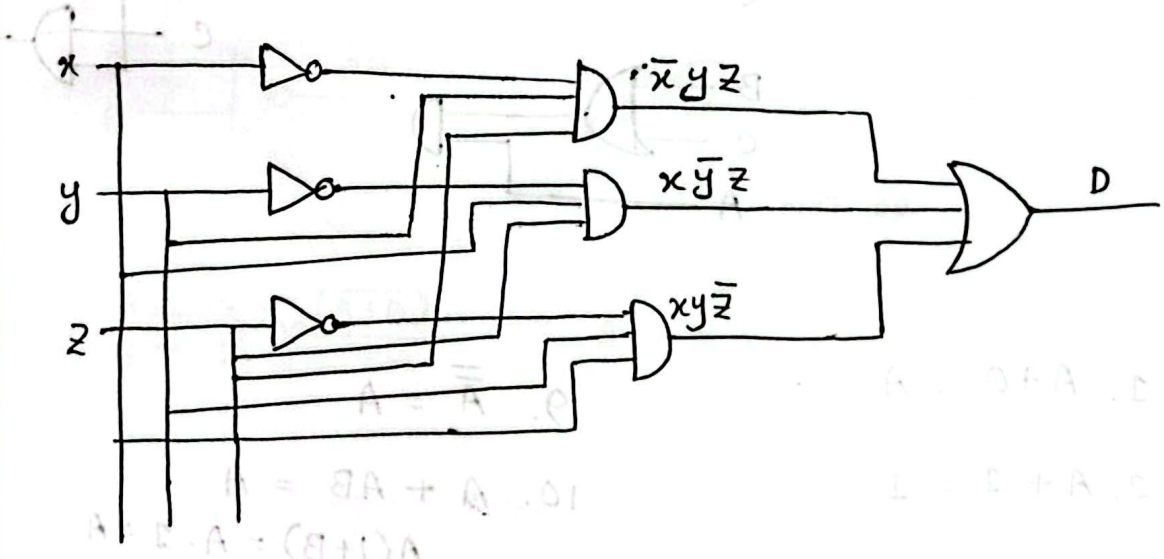
Digital waveform

x	y	z	D
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

যেকোনো দুইটা code 1 হলেই হবে।



$$D = \bar{x}yz + x\bar{y}z + xy\bar{z}$$



Boolean Algebra and Logic Simplification

Laws and Rules of Boolean Algebra

1. Commutative law

i) $A+B = B+A$

ii) $AB = BA$

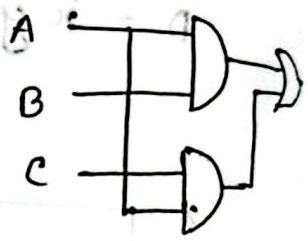
2. Associative law

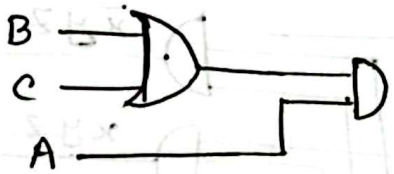
i) $A+(B+C) = (A+B)+C$

ii) $A(BC) = (AB)C$

3. Distributive law

$A(B+C) = AB+AC$





1. $A+0 = A$
2. $A+1 = 1$
3. $A \cdot 0 = 0$
4. $A \cdot 1 = A$
5. $A+A = A$
6. $A+\bar{A} = 1$
7. $A \cdot A = A$
8. $A \cdot \bar{A} = 0$
9. $\bar{\bar{A}} = A$
10. $A + AB = A$

    $A(1+B) = A \cdot 1 = A$
11. $A + \bar{A}B = A+B$

    $A + AB + \bar{A}B$

    $= A + B(A+\bar{A})$

    $= A + B \cdot 1$

    $= A + B$

CamScanner

12. $(A+B)(A+C) = A + BC$

$= AA + AC + AB + BC$

$= A + AC + AB + BC$

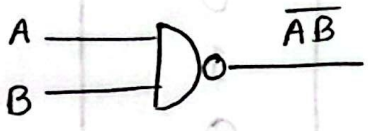
$= A(1 + C + B) + BC$

$= A \cdot 1 + BC = A + BC$

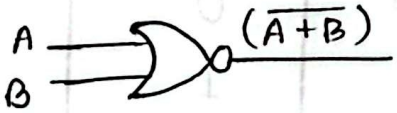
De Morgan's Laws :

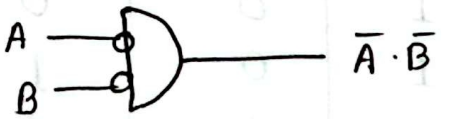
$\overline{AB} = \overline{A} + \overline{B}$ [NAND]

$\overline{(A+B)} = \overline{A}\,\overline{B}$ [NOR]



NOT OR





NOT AND

Simplification: Apply De Morgan's Law:

$$\overline{\overline{\overline{A + B\bar{C}} + D\,(\bar{E} + \bar{F})}}$$

$$= \overline{\overline{(A + B\bar{C})} \cdot \overline{(D\,(\bar{E} + \bar{F}))}}$$

$$= (A + B\bar{C})\,(\bar{D} + \overline{(\bar{E} + \bar{F})})$$

$$= (A + B\bar{C})\,(\bar{D} + E + F)$$

Draw A Truth table of $y = A\,(BC + D)$

A	B	C	D	BC	BC + D	$y = A\,(BC+D)$
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	0	0	0
0	0	1	1	0	1	0
0	1	0	0	0	0	0
0	1	0	1	0	1	0
0	1	1	0	1	1	0
0	1	1	1	1	1	0
1	0	0	0	0	0	0
1	0	0	1	0	1	1
1	0	1	0	0	0	0
1	0	1	1	0	1	1
1	1	0	0	0	0	0
1	1	0	1	0	1	1
1	1	1	0	1	1	1
1	1	1	1	1	1	1

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Simplify the following expression using boolean laws and rules:

$$[A\bar{B}(C+BD)+\bar{A}\bar{B}]C$$

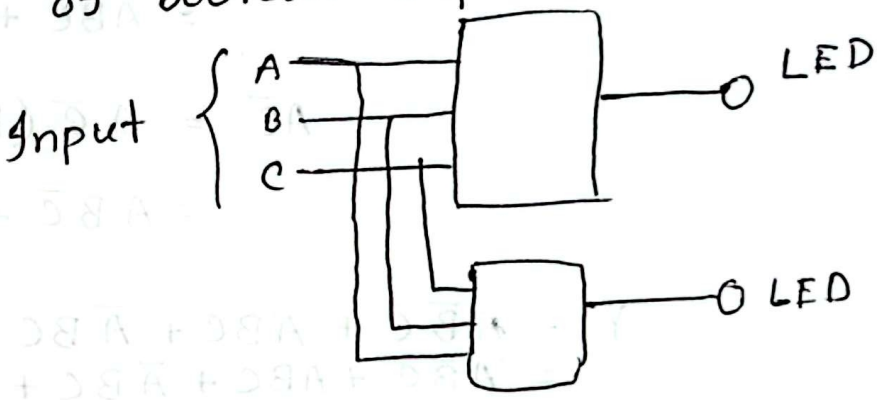
$$= [A\bar{B}C + A\underbrace{\bar{B}B}_{0}D + \bar{A}\bar{B}]C$$

$$= A\bar{B}C + \bar{A}\bar{B}C$$

$$= \bar{B}C\underbrace{(A+\bar{A})}_{1}$$

$$= \bar{B}C$$

Lab - 2: simplification of boolean expression.



Forms of Boolean Expression

2 forms :

Standard form
1. Sum of product (SOP)
2. Product of Sum (PoS)

boolean term

SoP : $Y = A\bar{B}\bar{C} + BC + A\bar{C}$ (1 0 0; Not standard)

PoS : $Y = (A + \bar{B} + \bar{C})(B + C)(A + \bar{C})$ (0; 1; 1)

Standard : $BC = BC \cdot (A + \bar{A})$

$= ABC + \bar{A}BC$

$A\bar{C} = A\bar{C}(B + \bar{B})$

$= AB\bar{C} + A\bar{B}\bar{C}$

$\therefore Y = A\bar{B}\bar{C} + ABC + \bar{A}BC + AB\bar{C} + A\bar{B}\bar{C}$

$= A\bar{B}\bar{C} + ABC + \bar{A}BC + AB\bar{C}$

using term value

$Y = A\bar{B}\bar{C} + BC + A\bar{C}$

$A\bar{B}\bar{C}$: 1 0 0

BC : 0 1 1, 1 1 1 (for A)

$A\bar{C}$: 1 0 0, 1 1 0 (for B)

$= A\bar{B}\bar{C} + \bar{A}BC + ABC + A\bar{B}\bar{C} + AB\bar{C}$

$= A\bar{B}\bar{C} + \bar{A}BC + ABC + AB\bar{C}$

Standard : $Y = (A + \bar{B} + \bar{C}) \cdot (B + C)(A + \bar{C})$

$B + C = (B + C + A\bar{A}) = (B + C + A)(B + C + \bar{A})$ (Distributive)

$A + C = (A + \bar{C} + B\bar{B}) = (A + \bar{C} + B)(A + \bar{C} + \bar{B})$

$\therefore Y = (A + \bar{B} + \bar{C})(A + B + C)(\bar{A} + B + C)(A + \bar{C} + B)(A + \bar{C} + \bar{B})$

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term value : $Y = (A + \bar{B} + \bar{C})(B + C)(A + \bar{C})$

$(A + \bar{B} + \bar{C})$ → 0 1 1

$(B + C)$ → 0 0 0 / 1 0 0 (A)

$(A + \bar{C})$ → 0 0 1 / 0 1 1 (for B)

[output 0] A → 0, B → 0, C → 0

$= (A + \bar{B} + \bar{C})(A + B + C)(\bar{A} + B + C)(A + B + \bar{C})$

$(A + \bar{B} + \bar{C})$

Converting (SoP ⟶ PoS)

$Y = A\bar{B}\bar{C} + BC + A\bar{C}$

output 1 { 1 0 0 → 4 ; 0 1 1 → 3 ; 1 0 0 → 4 ; 1 1 1 → 7 ; 1 1 0 → 6

Absent :

0	1	2	5
000	001	010	101

$(A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+\bar{C})$

POS

Truth table :

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

1 - SoP
0 - PoS

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absent

2 5 6 7

y = (PoS to SoP)

$Y = (A + \bar{B} + \bar{C})(B + C)(A + \bar{C})$

output { 0 1 1 → 3

0 0 0 → 0 0 0 1 → 1

1 0 0 → 4 0 1 1 → 3

0 1 0
$\bar{A}\,B\,\bar{C}$

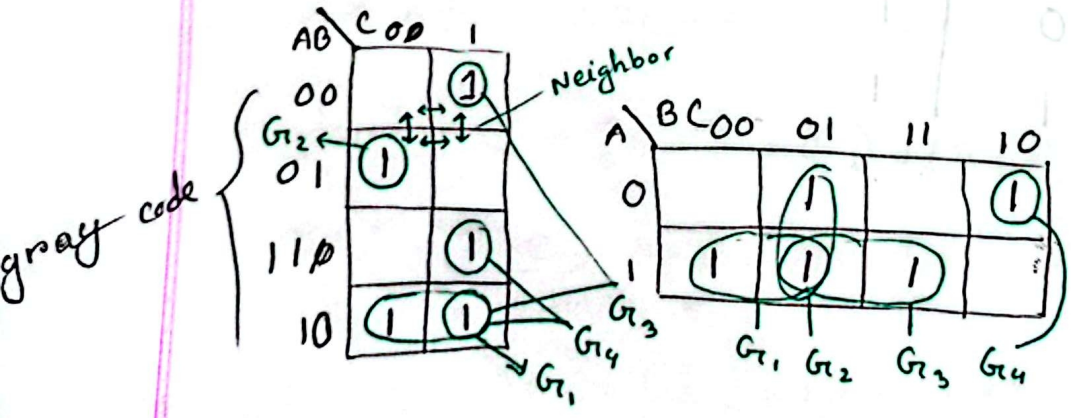
Truth table:

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

SoP : $\bar{A}B\bar{C} + A\bar{B}C + AB\bar{C} + ABC$

Simplification using Karnaugh Map
(Standard form required)

$Y = \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}B\bar{C} + ABC$] 1 for SoP



Grade Board

Unweighted binary code

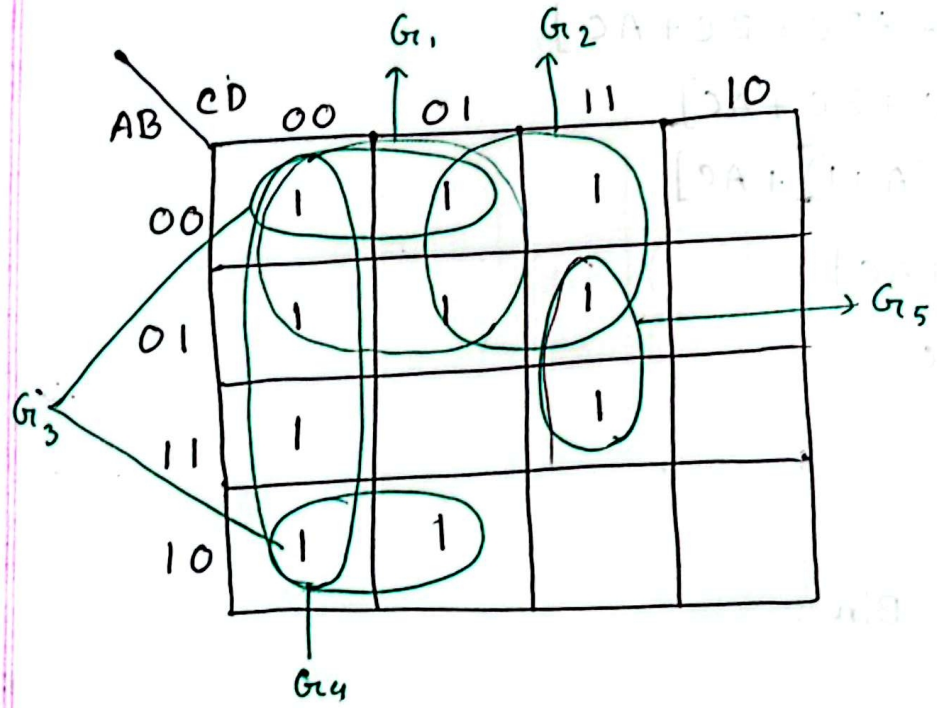
00
01
11
10
} Property of gray code

Grey Code

000 — 0
001 — 1
011 — 2
010 — 3
110 — 4
111 — 5
101 — 6
100 — 7

Minimized $Y : A\bar{B} + \bar{A}B\bar{C} + \bar{B}C + AC$

$$Y = A\bar{B}\bar{C}\bar{D} + AB\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}BCD + \bar{A}B\bar{C}D + A\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + ABCD$$



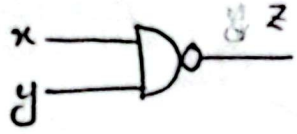
$$Y = \bar{A}\bar{C} + AD + \bar{B}\bar{C} + \bar{C}\bar{D} + BCD$$

CT - 1 : Wednesday

Universal Gate

NAND, NOR

NAND as universal gate :



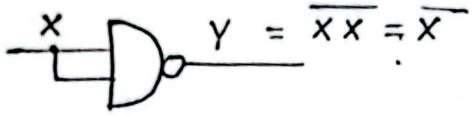
ONE NAND GATE → NOT (Inverter)

TWO NAND GATE → AND

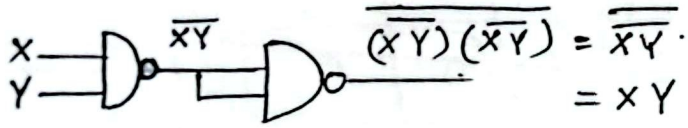
THREE NAND GATE → OR

FOUR NAND GATE → NOR

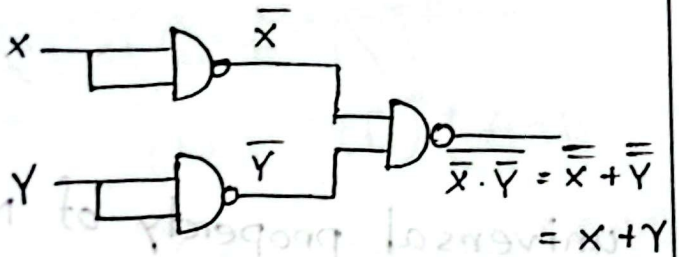
NOT :



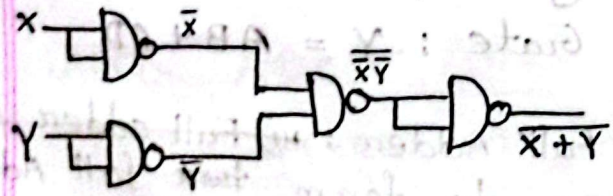
AND : $z = xy$



OR : $z = x + y$

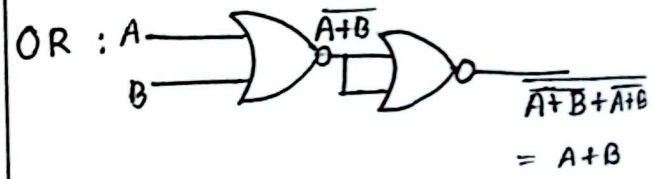


NAND → NOR :

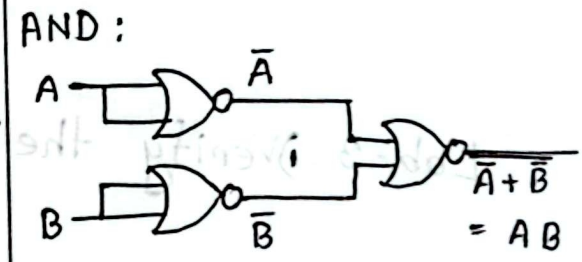


NOR as Universal Gate

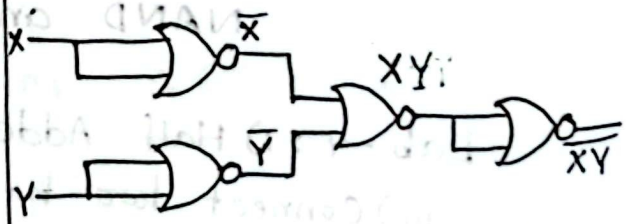
NOT : A → $\overline{A}$



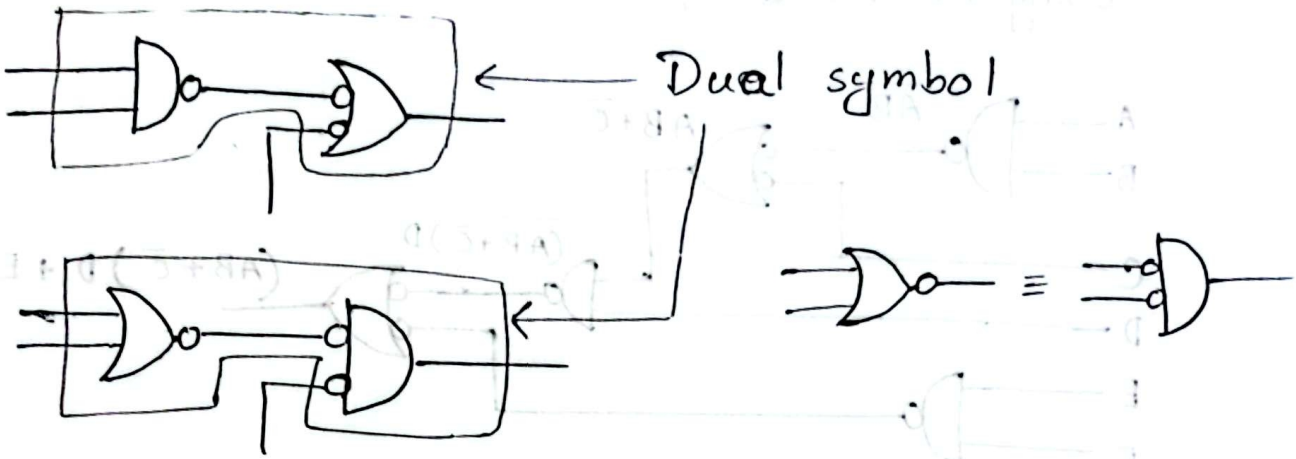
AND :



NOR → NAND :

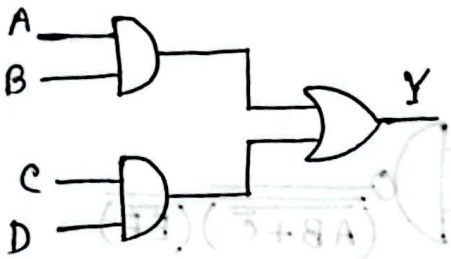


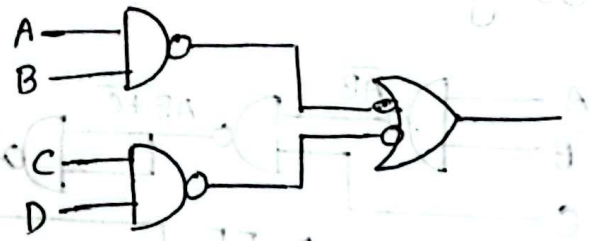
1 NOR → NOT (Inverter)
2 NOR → OR
3 NOR → AND
4 NOR → NAND



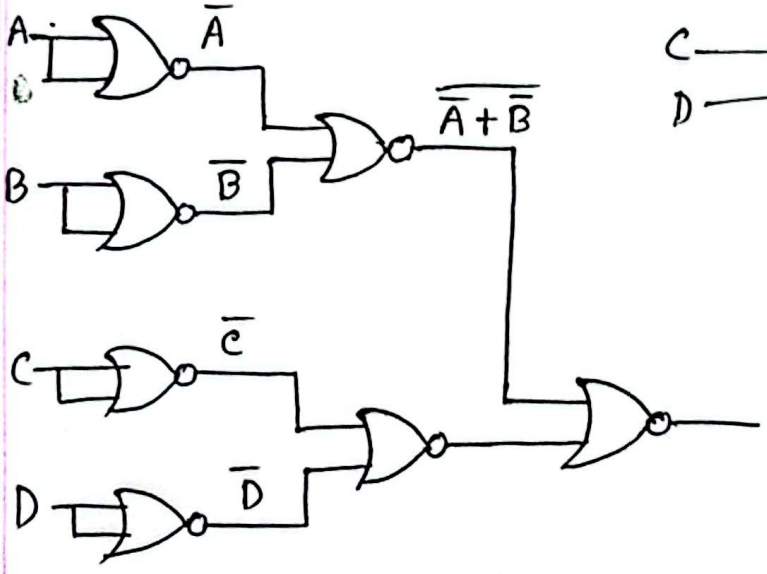
$Y = AB + CD$

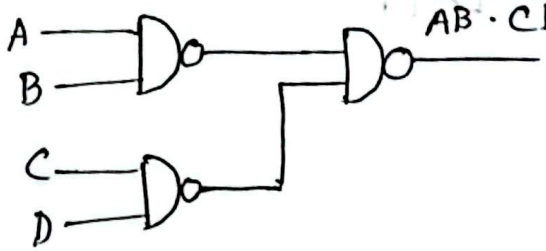
Using NAND & Dual symbol





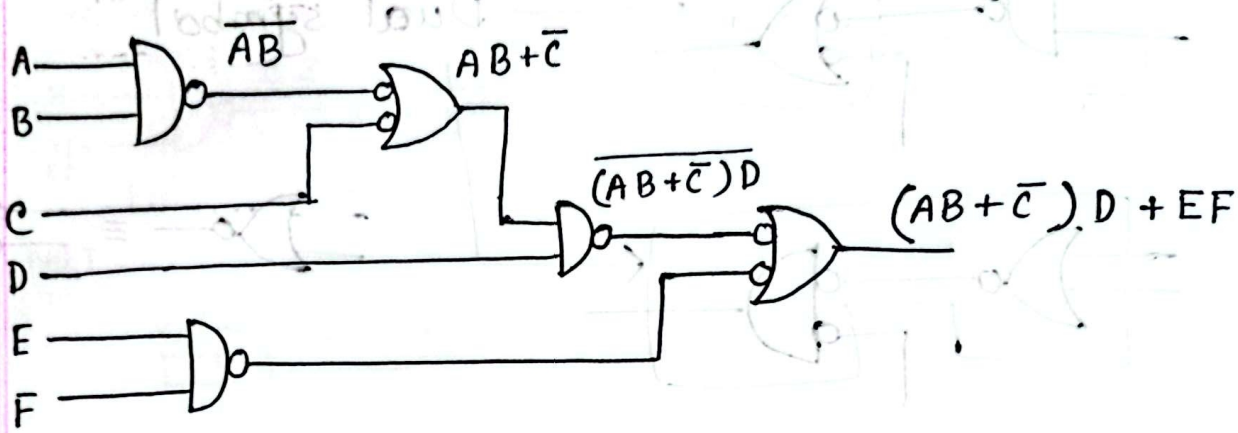
Using NOR gate



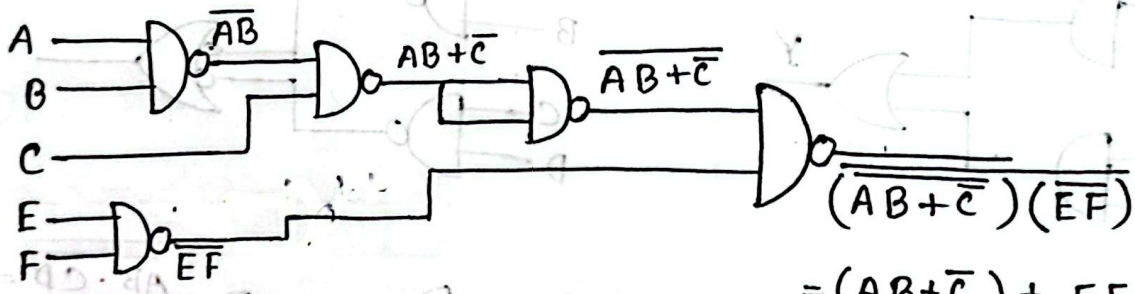


$Y = (AB + \bar{C})D + EF$

Using ~~NAND~~ Dual Symbol :



Using NAND :



$= (AB+\bar{C}) + EF$

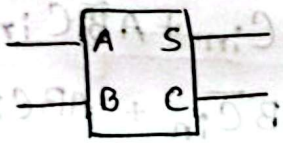
Functions of Combinational Logic

i) Addition : Adder < Half Adder (Add 2 number of 1 bit without carry)
1 bit এর 2 টা number add করে
Full Adder (Add 2 number of 1 bit with carry bit)

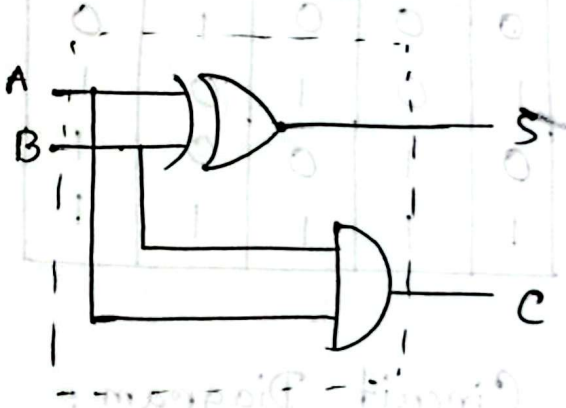
ii) Substruction: Substructor < Half Substructor
Full substructor

Half Adder
(2 input, 2 output)

Symbol :



Circuit Diagram



Truth Table :

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Expression (SOP)

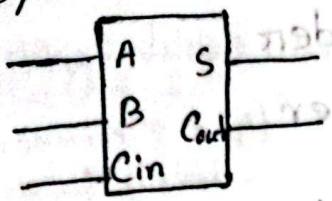
$S = \bar{A}B + A\bar{B} = A \oplus B$

$C = AB$

Full Adder:

(3 binary inputs, 2 outputs)

Symbol:



Truth Table:

A	B	Cin	S	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Expression (SOP):

$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$$

$$= C_{in}(\bar{A}\bar{B} + AB) + \bar{C}_{in}(\bar{A}B + A\bar{B})$$

$$= \overline{(A \oplus B)}C_{in} + (A \oplus B)\bar{C}_{in}$$

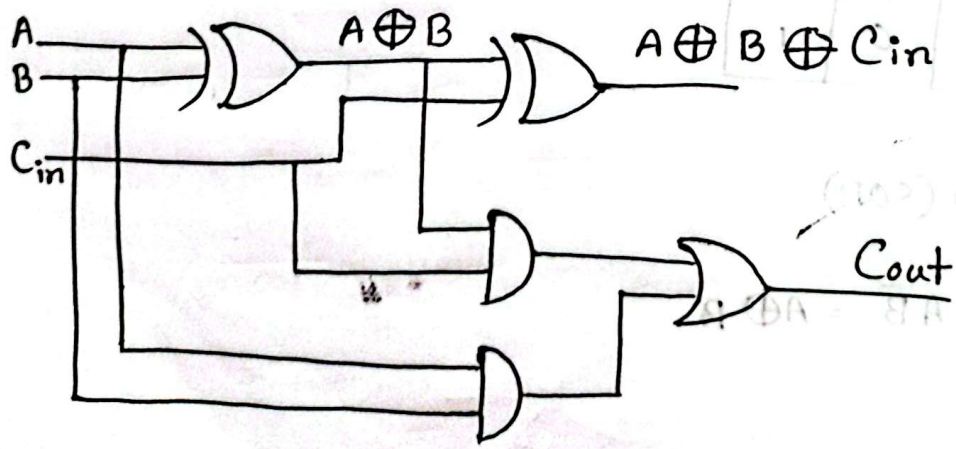
$$= A \oplus B \oplus C$$

$$C_{out} = \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in}$$

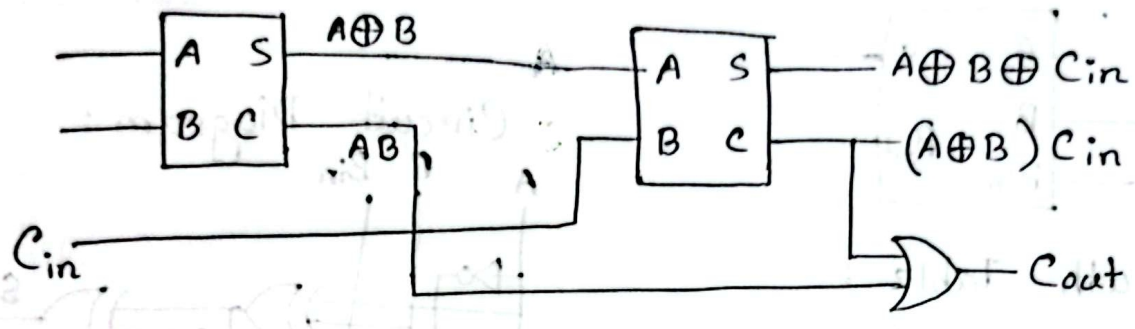
$$= C_{in}(\bar{A}B + A\bar{B}) + AB(C_{in} + \bar{C}_{in})$$

$$= (A \oplus B)C_{in} + AB$$

Circuit Diagram:

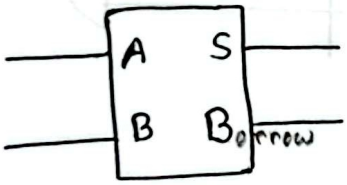


Full Adder with two half Adder (Block Diagram)



Half Substractor :

Symbol:



Expression (SOP) :

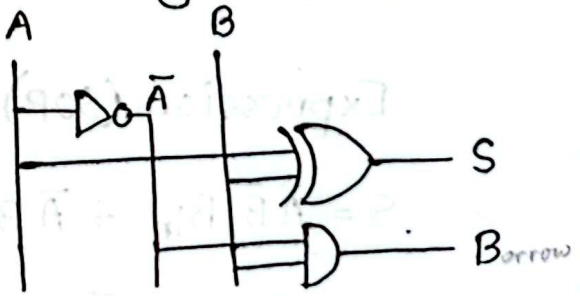
$$S = \bar{A}B + A\bar{B} = A \oplus B$$

$$B_{orrow} = \bar{A}B$$

Truth Table :

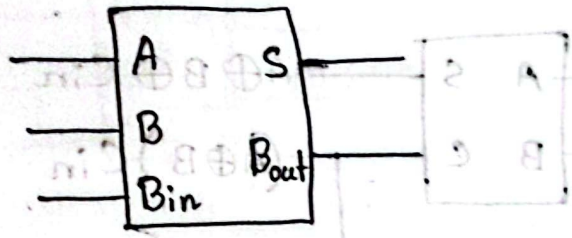
A	B	S	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Circuit Diagram :

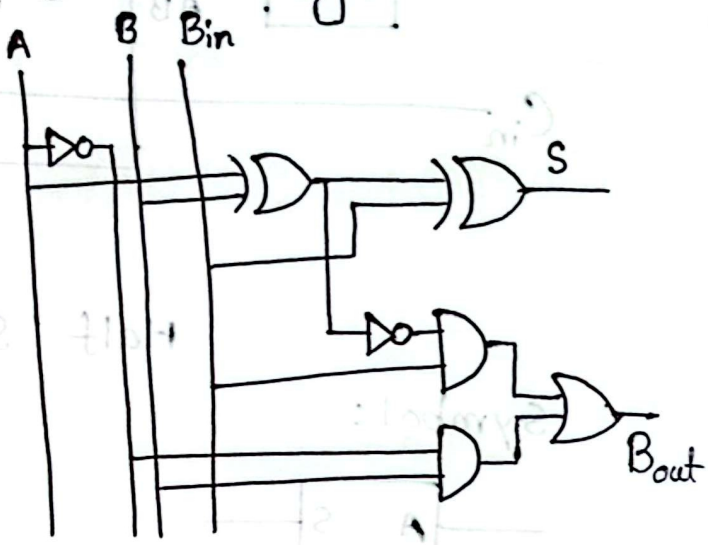


Full Substractor:

Symbol :



Circuit Diagram :



Truth Table :

A	B	Bin	S	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Expression (SOP) :

$$S = \bar{A}\bar{B}B_{in} + \bar{A}B\overline{B_{in}} + A\bar{B}\overline{B_{in}} + AB B_{in}$$

$$= B_{in}(\bar{A}\bar{B} + AB) + \overline{B_{in}}(\bar{A}B + A\bar{B})$$

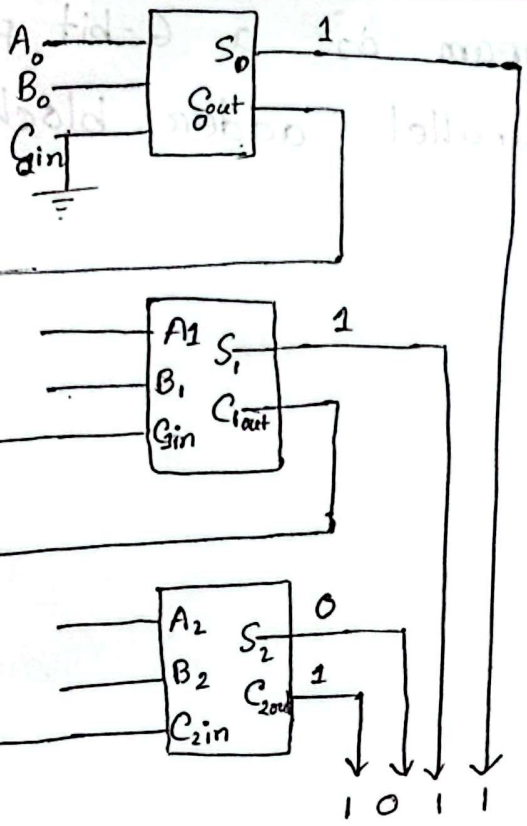
$$= B_{in}(\overline{A \oplus B}) + \overline{B_{in}}(A \oplus B)$$

$$= A \oplus B \oplus B_{in}$$

$$B_{out} = \bar{A}\bar{B}B_{in} + \bar{A}B\overline{B_{in}} + \bar{A}BB_{in} + ABB_{in}$$

$$= B_{in}(\bar{A}\bar{B} + AB) + \bar{A}B(B_{in} + \overline{B_{in}})$$

$$= B_{in}(\overline{A \oplus B}) + \bar{A}B$$

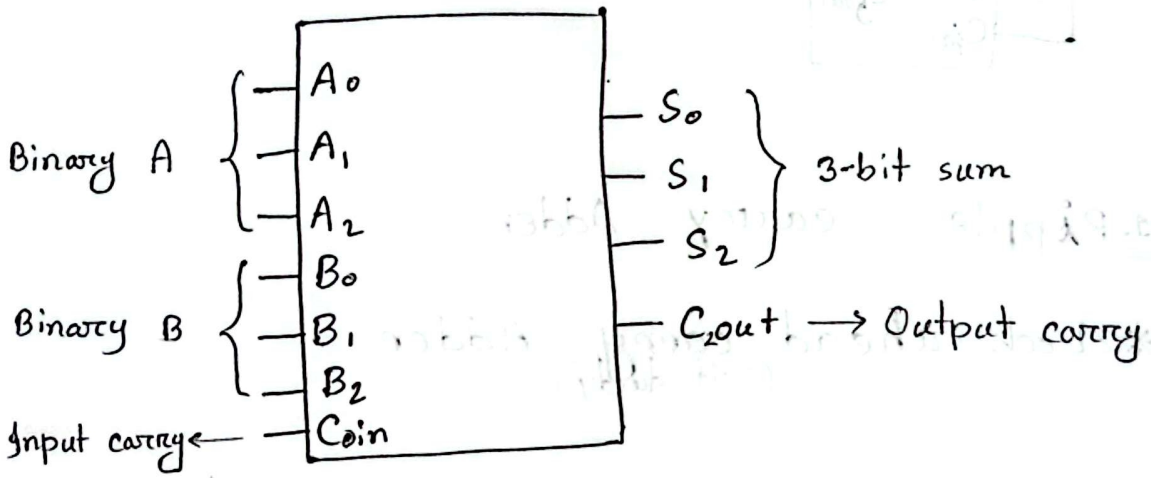


A = 101 $A_2 A_1 A_0$

B = 110 $B_2 B_1 B_0$

10 11

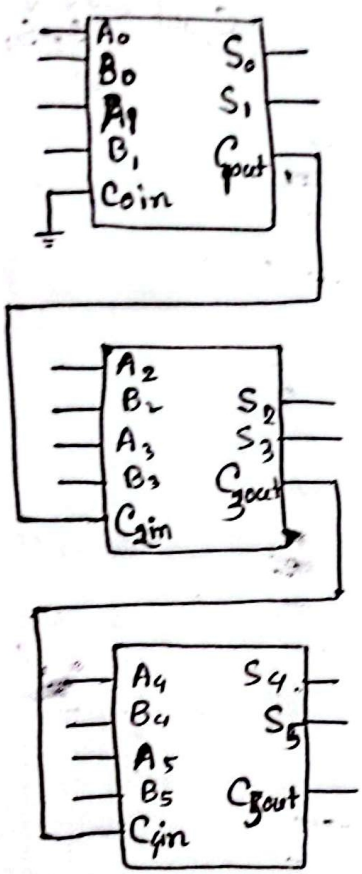
C_2out ← S_2 S_1 S_0



3-bit parallel Adder

(Block Diagram)

Draw the circuit diagram of a 6-bit parallel Adder using 2-bit parallel adder block.



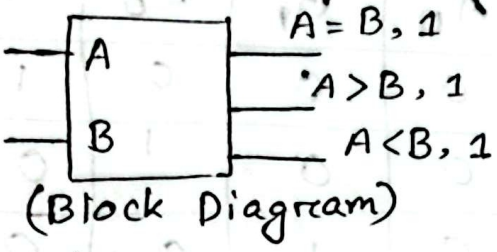
1. Ripple carry Adder

*2. Look ahead carry Adder
(self study)

Comparison Function

(Comparator): XNOR gate is a premitive/basic comparetor.

1 bit Comparator:



(Block Diagram)

[Only one will result = 1, others zero]

Truth Table:

A	B	X	Y	Z
0	0	Ø1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0

SOP. expression:

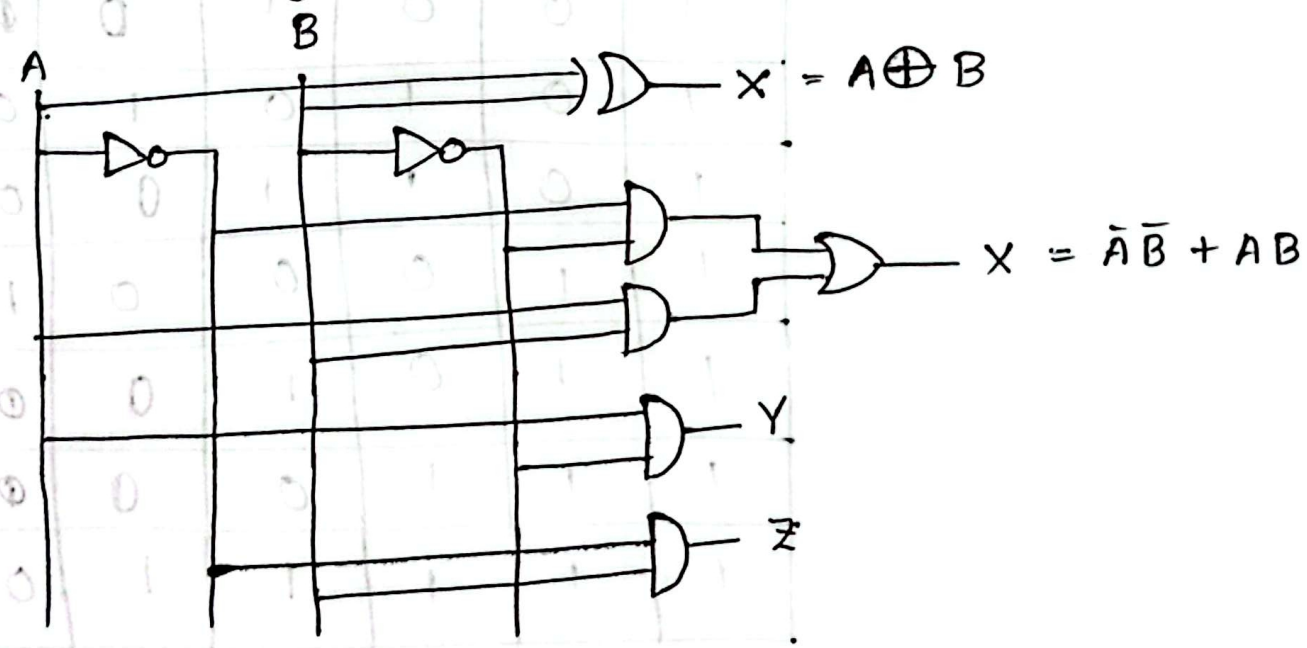
$X = \bar{A}\bar{B} + AB$

$= \overline{A \oplus B}$

$Y = A\bar{B}$

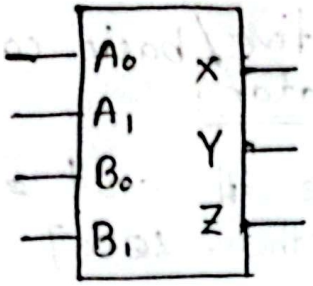
$Z = \bar{A}B$

Circuit Diagram:



1-bit comparator

2 bit Comparator



Block Diagram

X : A = B ; 1

Y : A > B ; 1

Z : A < B ; 1

Truth Table :

A_1	A_0	B_1	B_0	X	Y	Z
0	0	0	0	1	0	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	0	1	0
0	1	0	1	1	0	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	0	1	0
1	0	0	1	0	1	0
1	0	1	0	1	0	0
1	0	1	1	0	0	1
1	1	0	0	0	1	0
1	1	0	1	0	1	0
1	1	1	0	0	1	0
1	1	1	1	1	0	0

If, $A_1 = B_1$ AND $A_0 = B_0$ Then $X = 1$

If $A_1 > B_1$ OR $A_1 = B_1$ and $A_0 > B_0$ then, $A > B$
$Y = 1$

If $A_1 < B_1$ OR $A_1 = B_1$ and $A_0 < B_0$ then $Z = 1$

$$X = (\overline{A_1 \oplus B_1})(\overline{A_0 \oplus B_0})$$

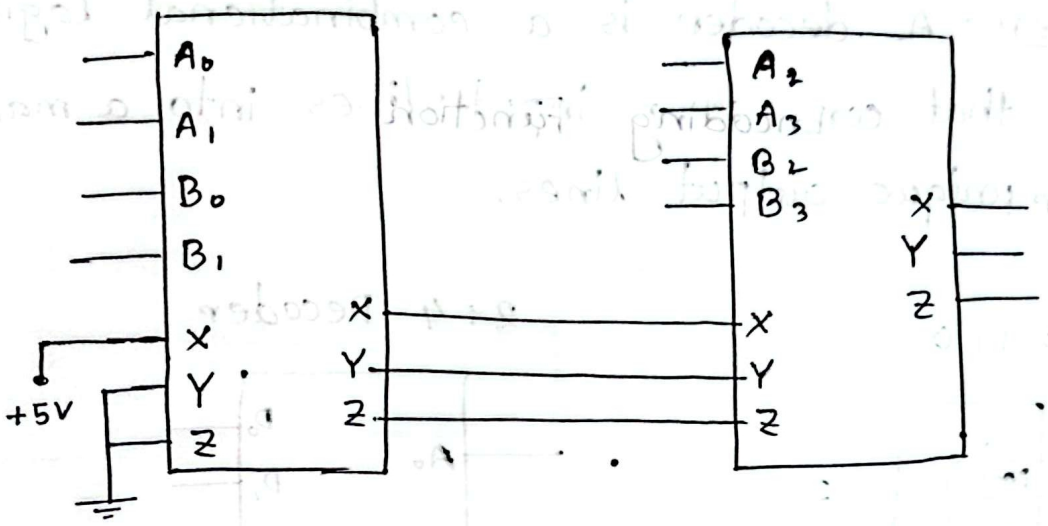
$$Y = A_1 \overline{B_1} + (\overline{A_1 \oplus B_1})\, A_0 \overline{B_0}$$

$$Z = \overline{A_1}\, B_1 + (\overline{A_1 \oplus B_1})\, \overline{A_0}\, B_0$$

CamScanner

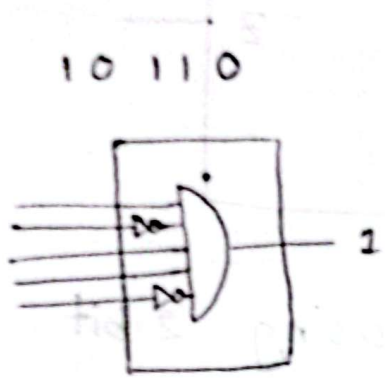
$A = A_3 A_2 A_1 A_0$

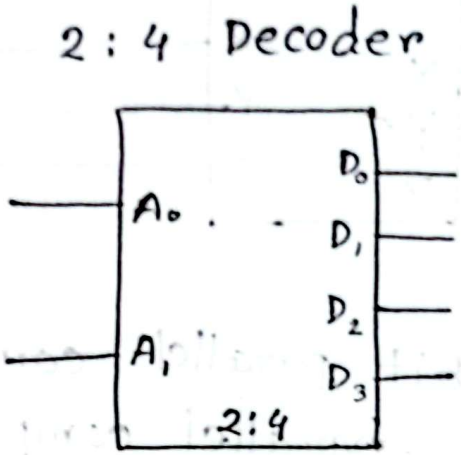
$B = B_3 B_2 B_1 B_0$



4-bit parallel comparator using 2-bit parallel comparator blocks.

Decoder : A decoder is a combinational logic circuit that converts n input lines into a maximum of 2^n unique output lines.





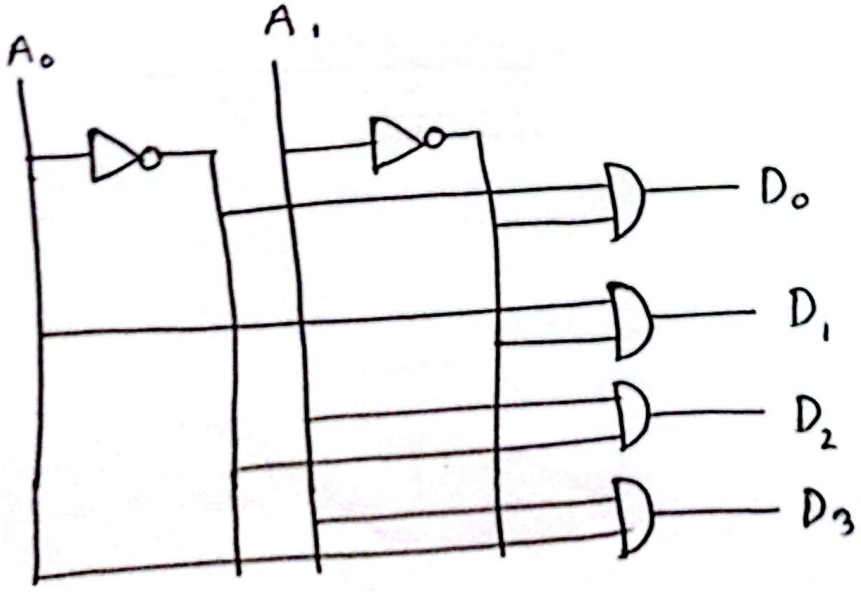
		D_3	D_2	D_1	D_0
A_1	A_0	D_0	D_1	D_2	D_3
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

$D_0 = \bar{A}_1 \bar{A}_0$

$D_1 = \bar{A}_1 A_0$

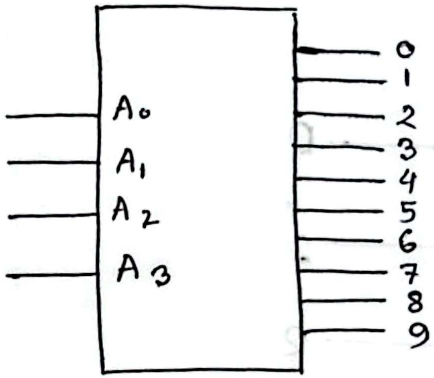
$D_2 = A_1 \bar{A}_0$

$D_3 = A_1 A_0$



4-to-10 Line Decoder

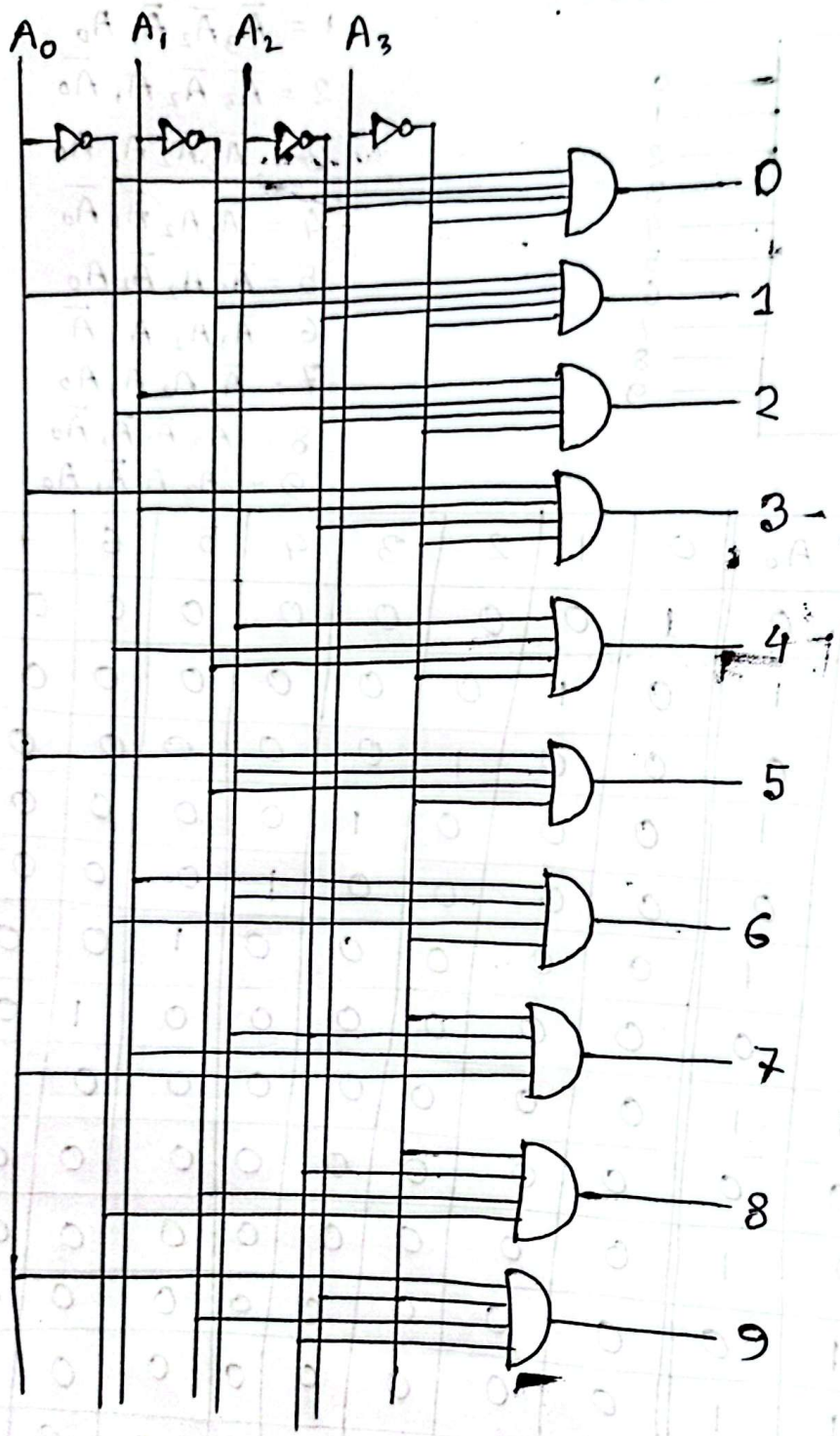
BCD to Decimal Decoder:



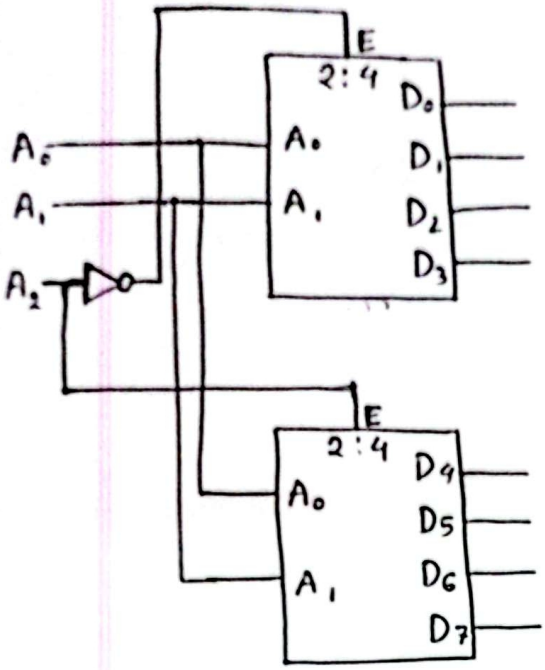
$0 = \bar{A}_3 \bar{A}_2 \bar{A}_1 \bar{A}_0$
$1 = \bar{A}_3 \bar{A}_2 \bar{A}_1 A_0$
$2 = \bar{A}_3 \bar{A}_2 A_1 \bar{A}_0$
$3 = \bar{A}_3 \bar{A}_2 A_1 A_0$
$4 = \bar{A}_3 A_2 \bar{A}_1 \bar{A}_0$
$5 = \bar{A}_3 A_2 \bar{A}_1 A_0$
$6 = \bar{A}_3 A_2 A_1 \bar{A}$
$7 = \bar{A}_3 A_2 A_1 A_0$
$8 = A_3 \bar{A}_2 \bar{A}_1 \bar{A}_0$
$9 = A_3 \bar{A}_2 \bar{A}_1 A_0$.

A_3	A_2	A_1	A_0	0	1	2	3	4	5	6	7	8	9
0	0	0	0	1	0	0	0	0	0	0	0	0	0
0	0	0	1	0	1	0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0	0	0	0	0	0	0
0	0	1	1	0	0	0	1	0	0	0	0	0	0
0	1	0	0	0	0	0	0	1	0	0	0	0	0
0	1	0	1	0	0	0	0	0	1	0	0	0	0
0	1	1	0	0	0	0	0	0	0	1	0	0	0
0	1	1	1	0	0	0	0	0	0	0	1	0	0
1	0	0	0	0	0	0	0	0	0	0	0	1	0
1	0	0	1	0	0	0	0	0	0	0	0	0	1

CamScanner

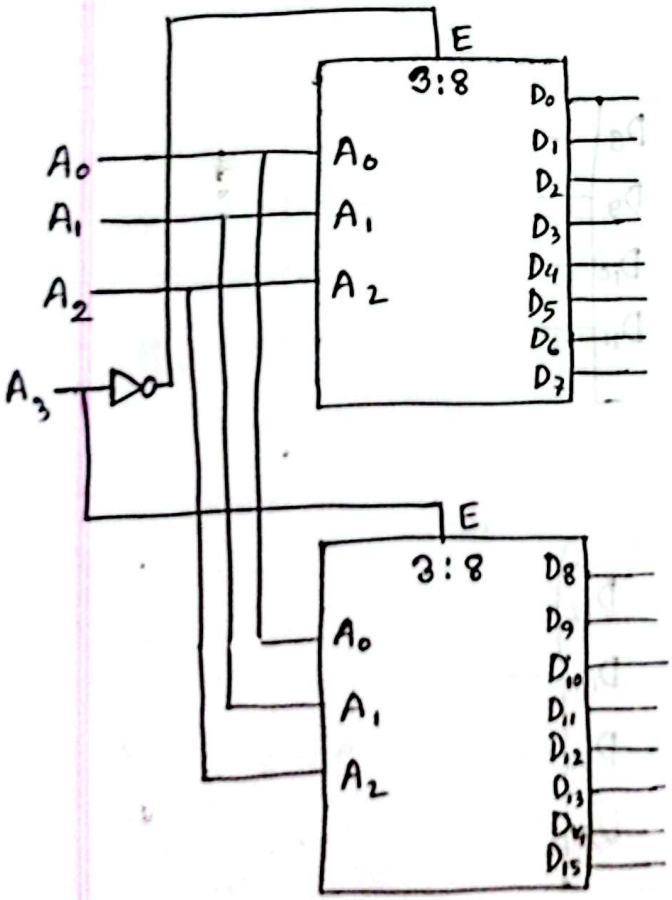

A0
A1
A2
A3
0
1
2
3
4
5
6
7
8
9

2 : 4 ⟶ 3 : 8 Decoder

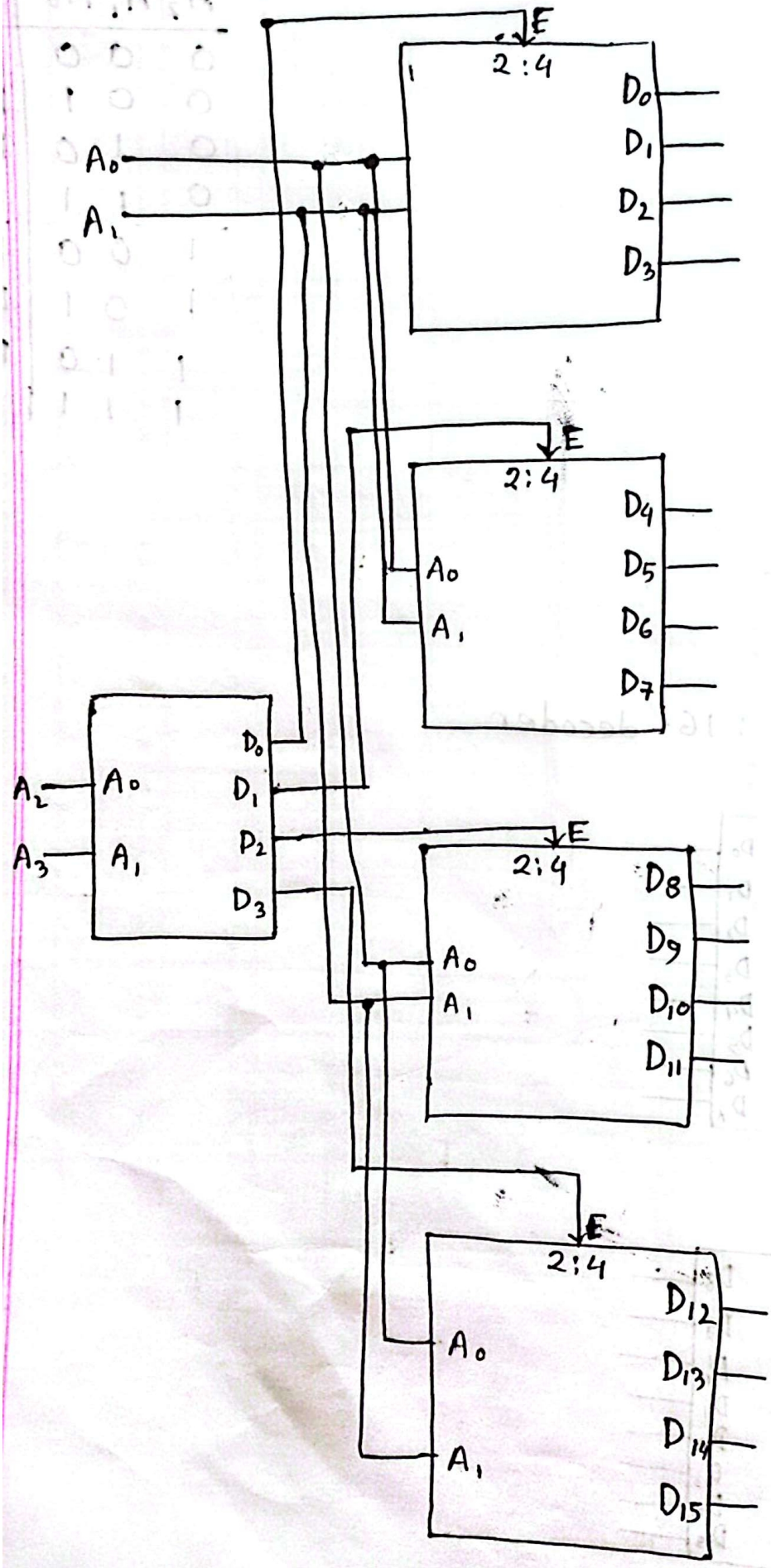


A_2	A_1	A_0	
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

3 : 8 ⟶ 4 : 16 decoder



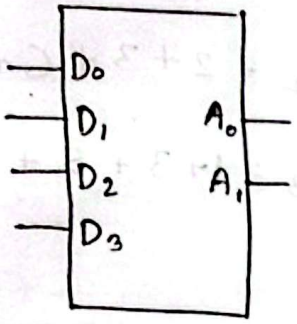
2:4 ⟶ 4:16 decoder



Encoding Function

Encoder :

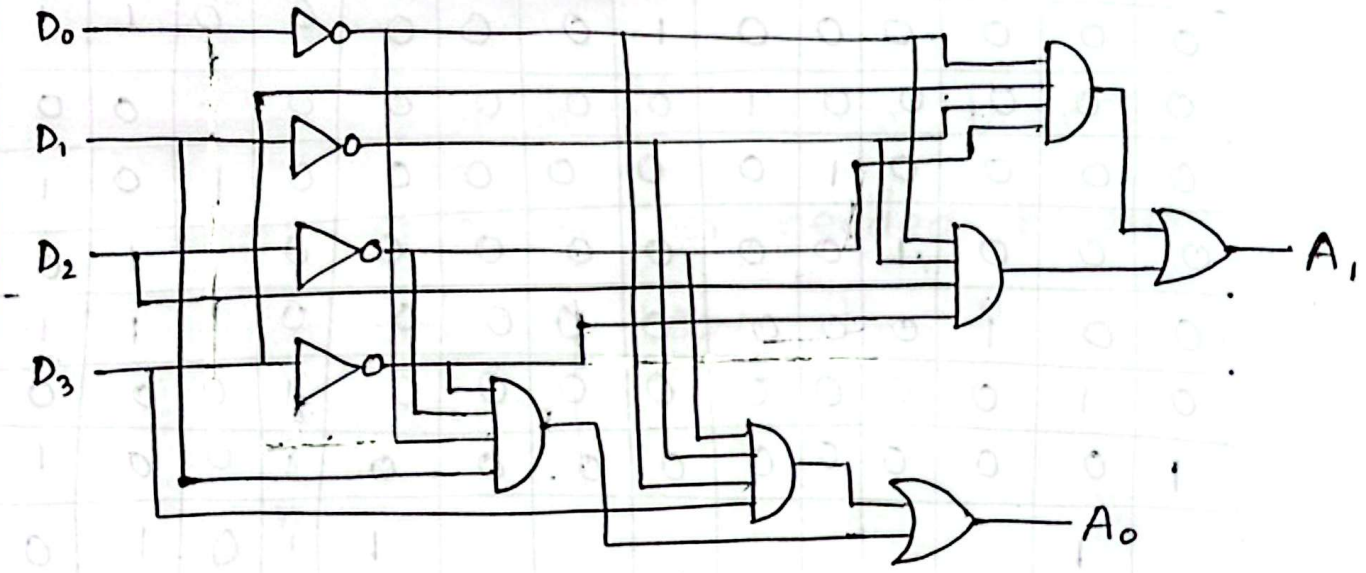
4 : 2



D_3	D_2	D_1	D_0	A_0	A_1
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

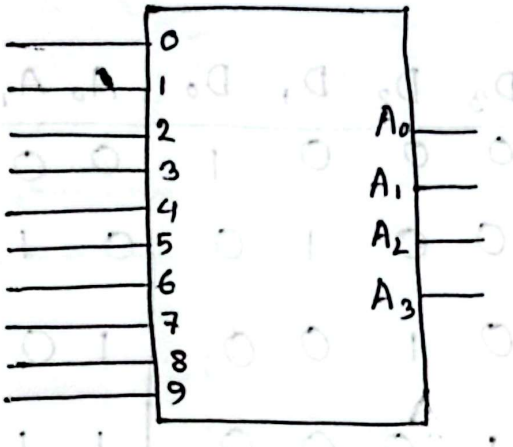
$$A_0 = \bar{D}_3 D_2 \bar{D}_1 \bar{D}_0 + D_3 \bar{D}_2 \bar{D}_1 \bar{D}_0$$

$$A_1 = \bar{D}_3 \bar{D}_2 D_1 \bar{D}_0 + D_3 \bar{D}_2 \bar{D}_1 \bar{D}_0$$



An encoder is a combinational logic circuit that perform the inverse operation of a decoder. It takes 2^n or fewer input lines and encodes them into an n-bit binary output.

Decimal to BCD Encoder:



$A_0 = 8 + 9$

$A_1 = 4 + 5 + 6 + 7$

$A_2 = 2 + 3 + 6 + 7$

$A_3 = 1 + 3 + 5 + 7 + 9$

9	8	7	6	5	4	3	2	1	0	A_0	A_1	A_2	A_3
0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	0	0	0	0	1	0	0	0	0	1	0
0	0	0	0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	0	1	0	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0	0	0	0	1	0	1
0	0	0	1	0	0	0	0	0	0	0	1	1	0
0	0	1	0	0	0	0	0	0	0	0	1	1	1
0	1	0	0	0	0	0	0	0	0	1	0	0	0
1	0	0	0	0	0	0	0	0	0	1	0	0	1

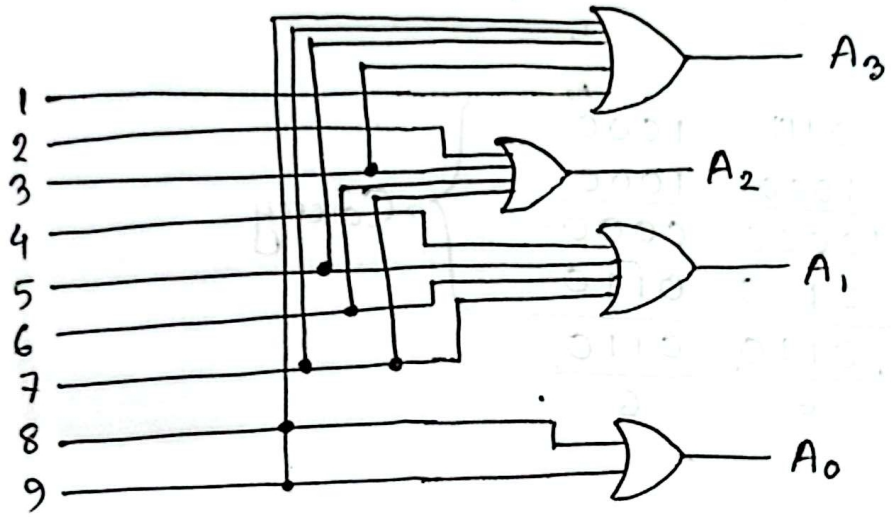
CamScanner

$A_0 = 8 + 9$

$A_1 = 4 + 5 + 6 + 7$

$A_2 = 2 + 3 + 6 + 7$

$A_3 = 1 + 3 + 5 + 7 + 9$

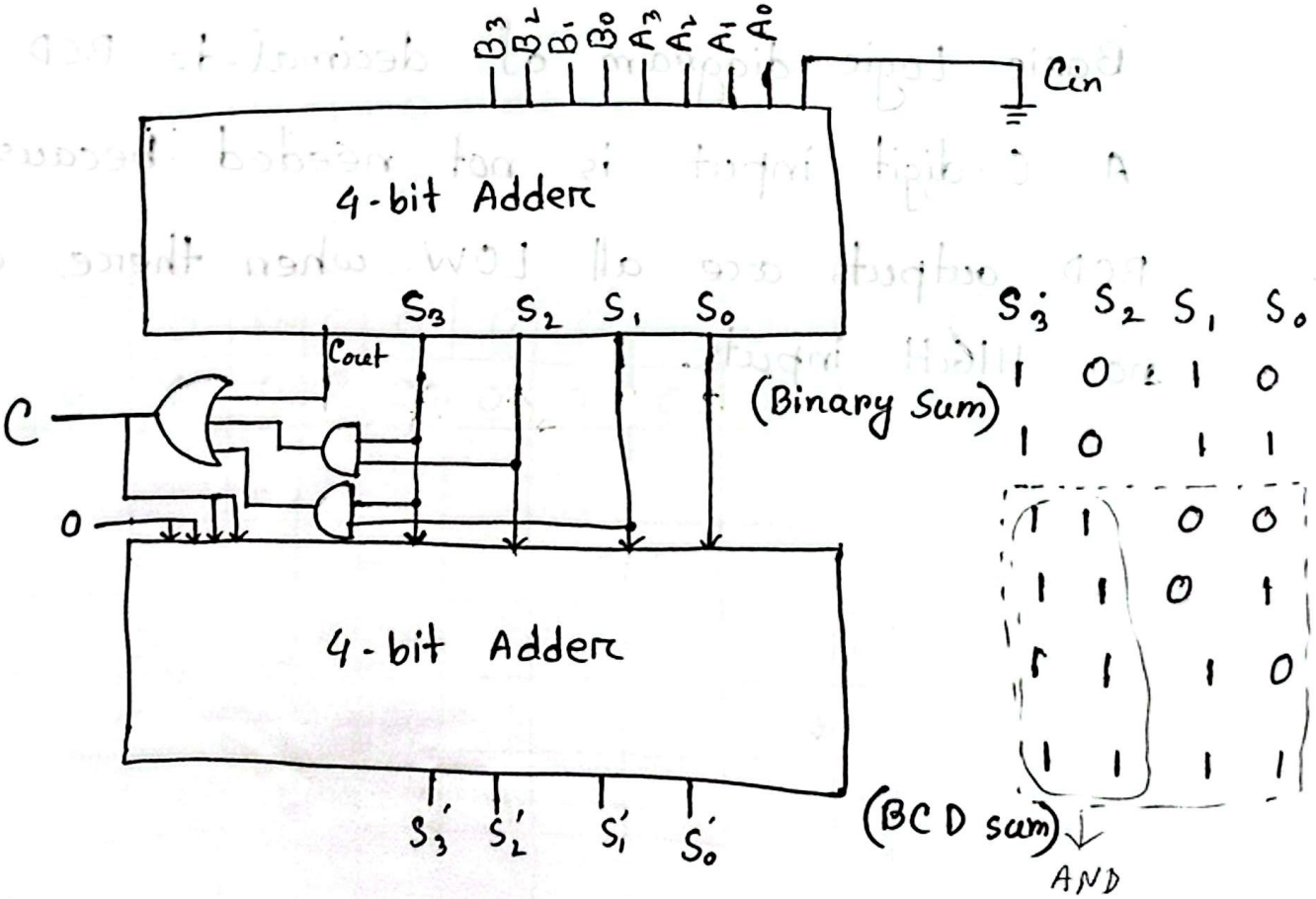


Basic Logic diagram of decimal-to-BCD encoder. A 0-digit input is not needed because the BCD outputs are all LOW when there are no HIGH inputs.

BCD Adder

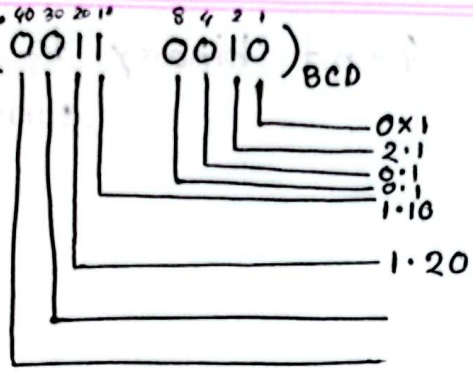
```
78          0111  1000        Carry or above 9
87          1000  0111
165         1111  1111  ---> above 9 add 6
            0110  0110  <-6
          1 0110  0101
            1     5
            6
```

```
78          A3A2A1A0
88          0111  1000
166         1000  1000
          10000  0000  ---> Carry
            0110  0110
          1 0110  0110
            1     6
            6
```



S_3	S_2	S_1	S_0
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

$(32)_{10}$ = $(0011\ 0010)_{BCD}$




```
  0010
  1010
 10100
------
100000
```

[Assignment on BCD to Binary conversion +Viva
Input → BCD
Output → Binary
Don't use function (build in); in C/C++]

Binary → Gray
↓ positional value আছে
↓ positional value নাই

(B to G) B → 1011 1100
G → 1110 1010

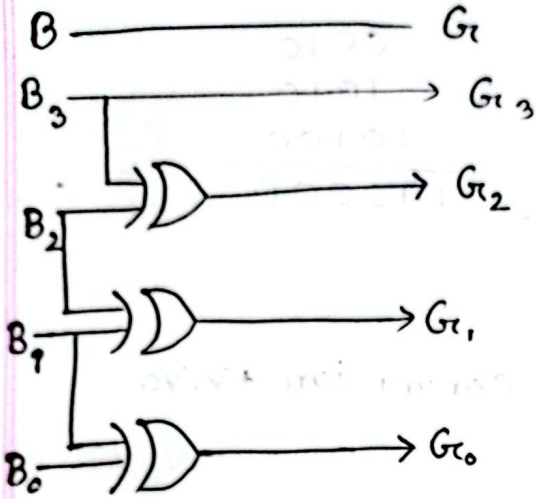
OR

↓ most significant bit will same for both

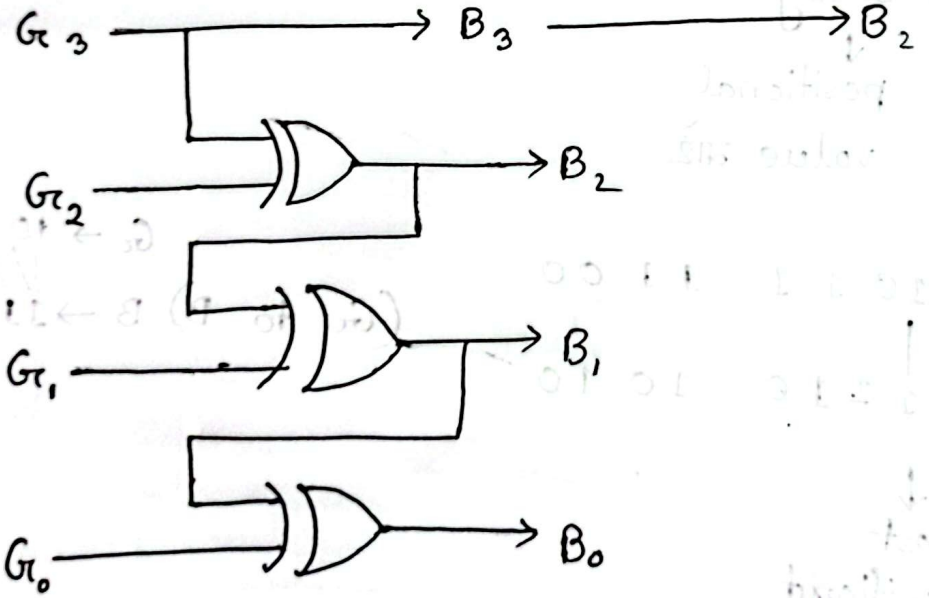
(G to B) G → 1011 1100
B → 1101 1000

XOR

4-bit binary to gray code converter



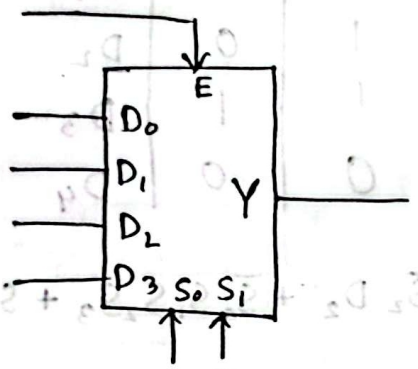
4-bit Gray Code to Binary Converter



Multiplexer / Data Selector

(একই পথ দিয়ে অনেক কিছু যেতে পারে)

4 : 1 MUX



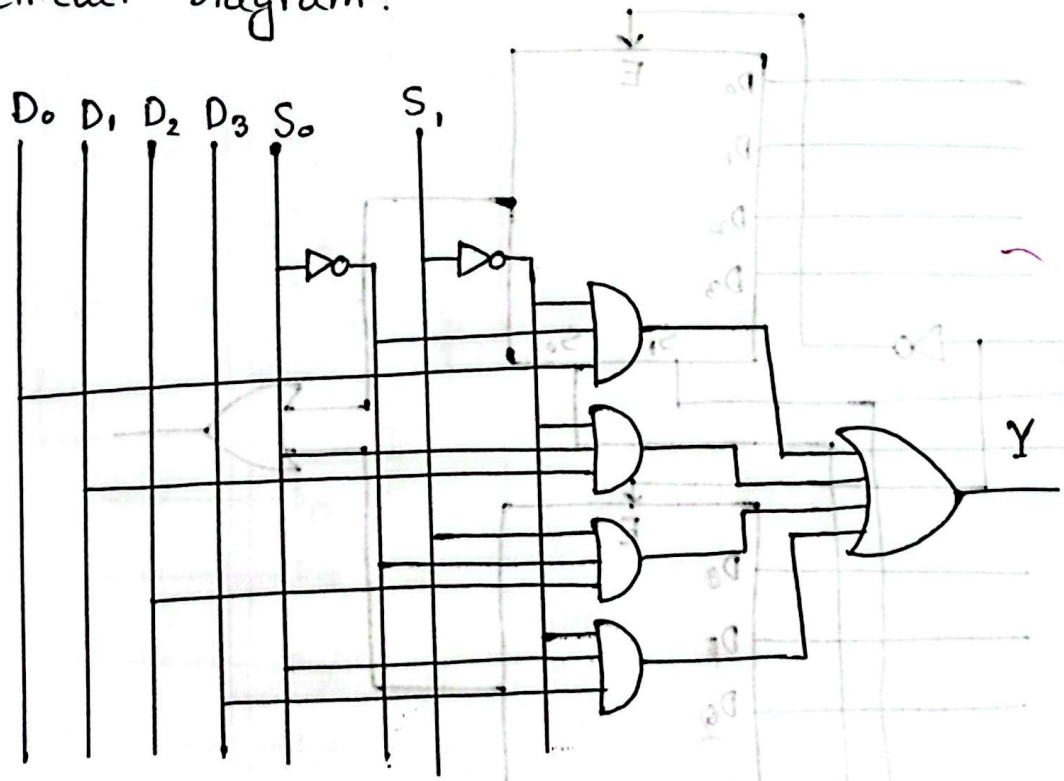
Enable input

Truth table

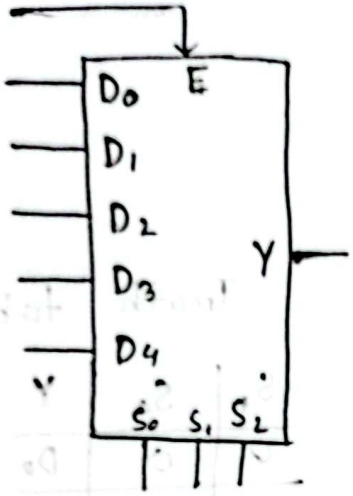
S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

$$Y = \bar{S}_1 \bar{S}_0 D_0 + \bar{S}_1 S_0 D_1 + S_1 \bar{S}_0 D_2 + S_1 S_0 D_3$$

Circuit Diagram:



5:1 MUX

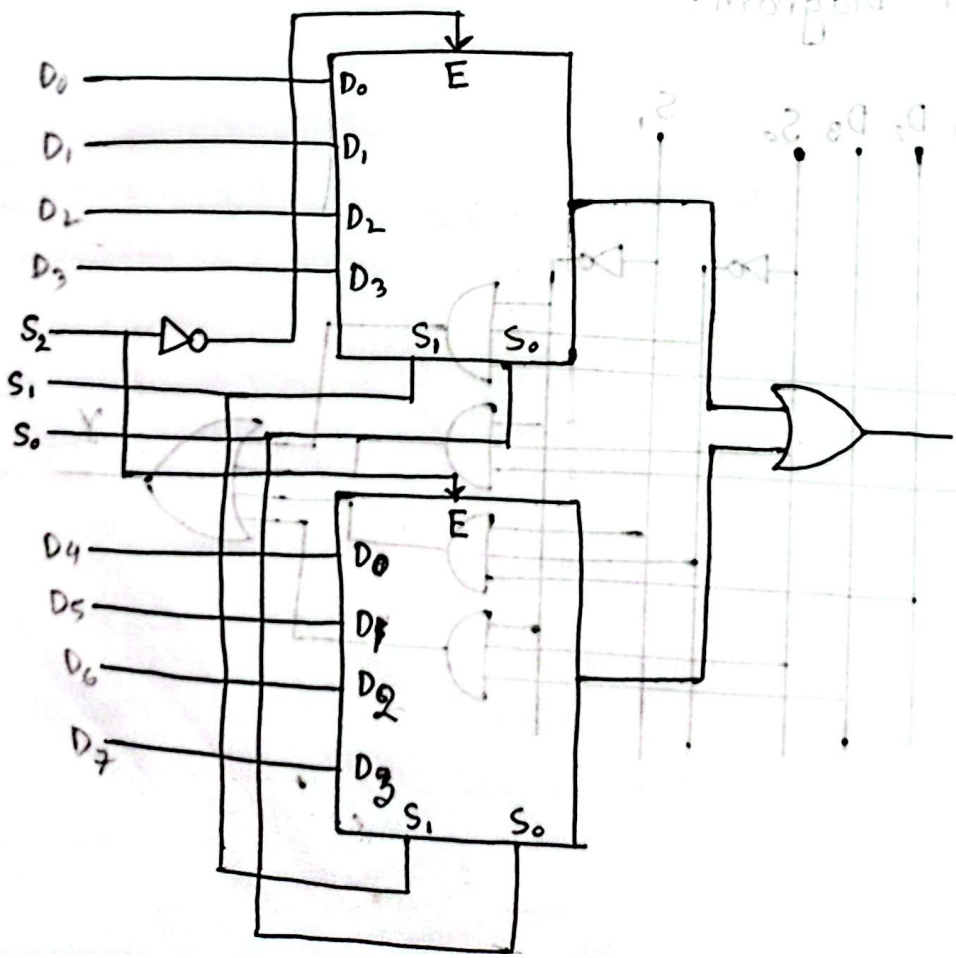


Truth Table:

S_2	S_1	S_0	Y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4

$$Y = \bar{S}_0 \bar{S}_1 \bar{S}_2 D_0 + \bar{S}_0 \bar{S}_1 S_2 D_1 + \bar{S}_0 S_1 \bar{S}_2 D_2 + \bar{S}_0 S_1 S_2 S_3 + S_0 \bar{S}_1 \bar{S}_2 D_4$$

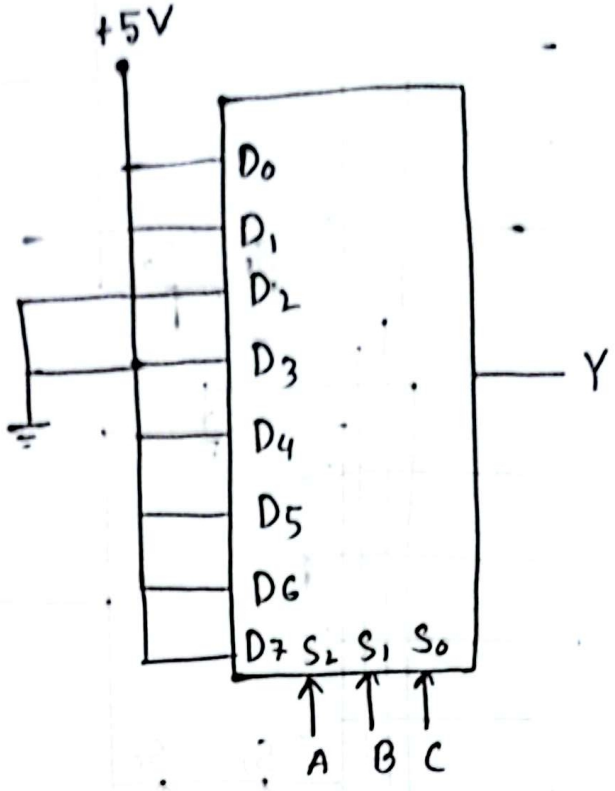
Convert 4:1 to 8:1 MUX

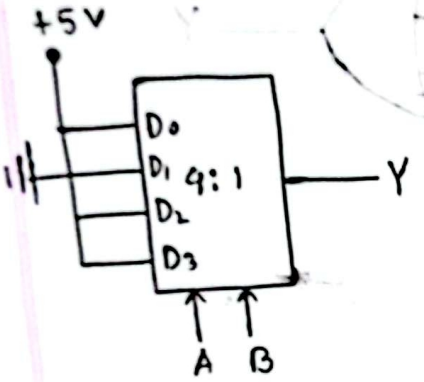


$Y = \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + AB\bar{C} + \bar{A}\bar{B}C + A\bar{B}C + ABC$

Implement the following exp. using 4:1 MUX

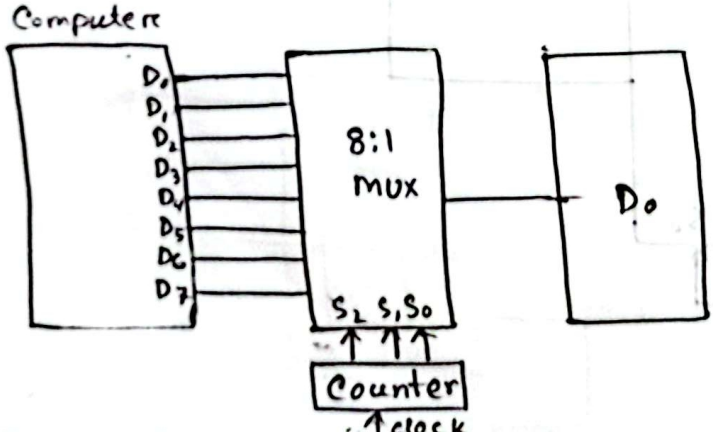
A	B	C	Y
0	0	0	1
0	0	1	1
0	0	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1





Application of MUX

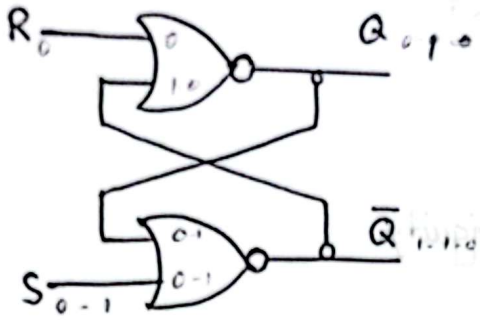
(Parallel to Serial data Converter)



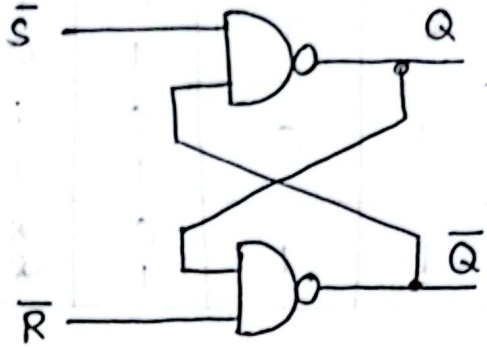
Latches & Flip-flops (Sequential logic)

DeMultiplexer (self study

S-R Latch



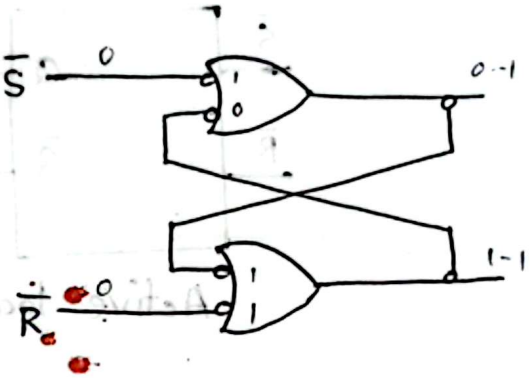
Active high Input



Active low Input

1 - high
0 - low

≡



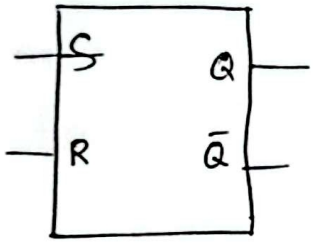
Truth Table:

$\bar{S}$	$\bar{R}$	Q^{th}	Q	
0	0	0	*	ambiguity
		1	*	
0	1	0	1	Set
		1	1	
1	0	0	0	Reset
		1	0	
1	1	0	0	No change
		1	1	

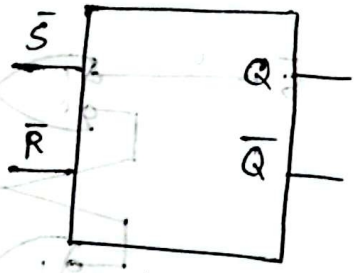
CamScanner

Active high Input's Truth Table

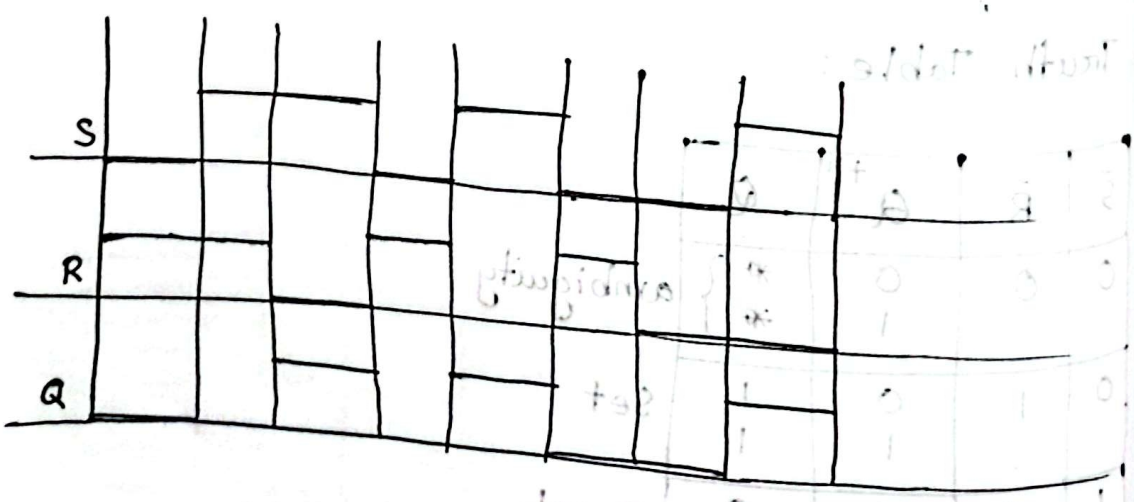
S	R	Q^{+1}	Q^{+}	
0	0	0	0	No Change
		1	0	
0	1	0	0	Reset
		1	0	
1	0	0	1	Set
		1	1	
1	1	0	*	ambiguity
		1	*	

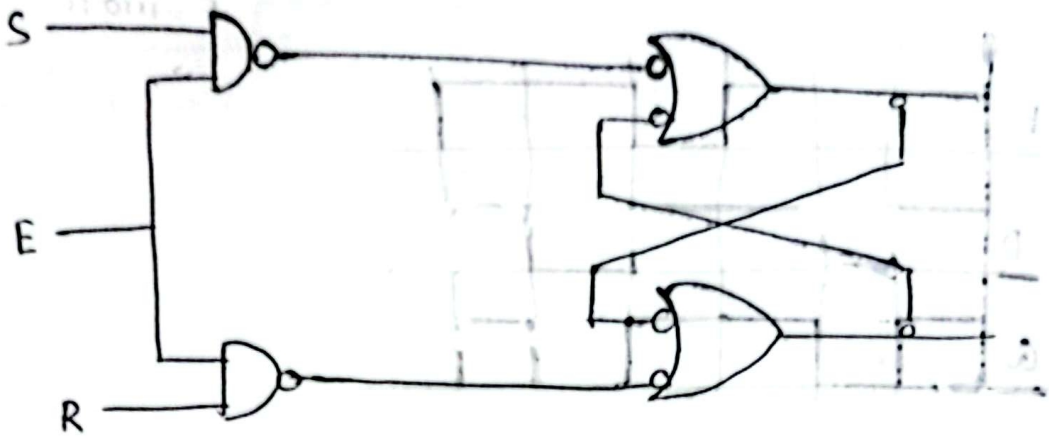


Active High

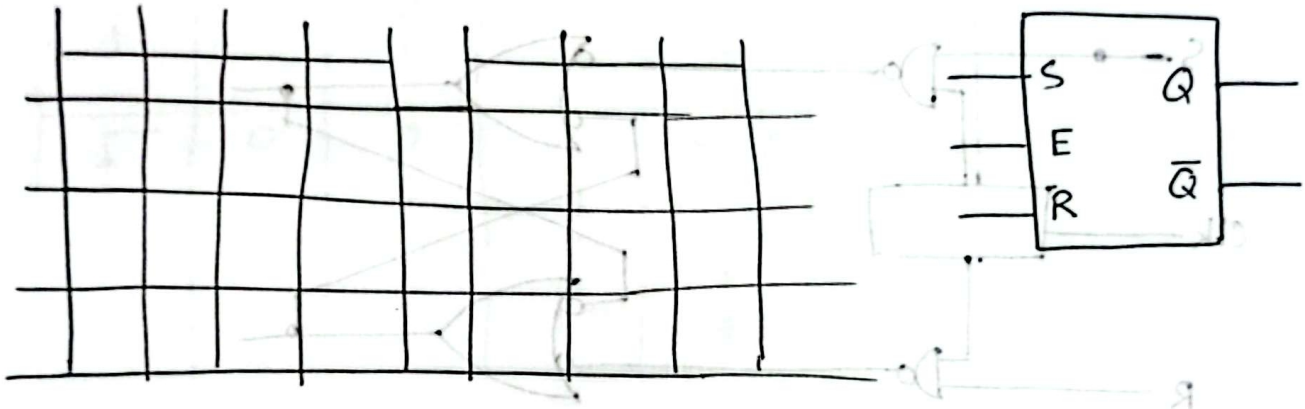


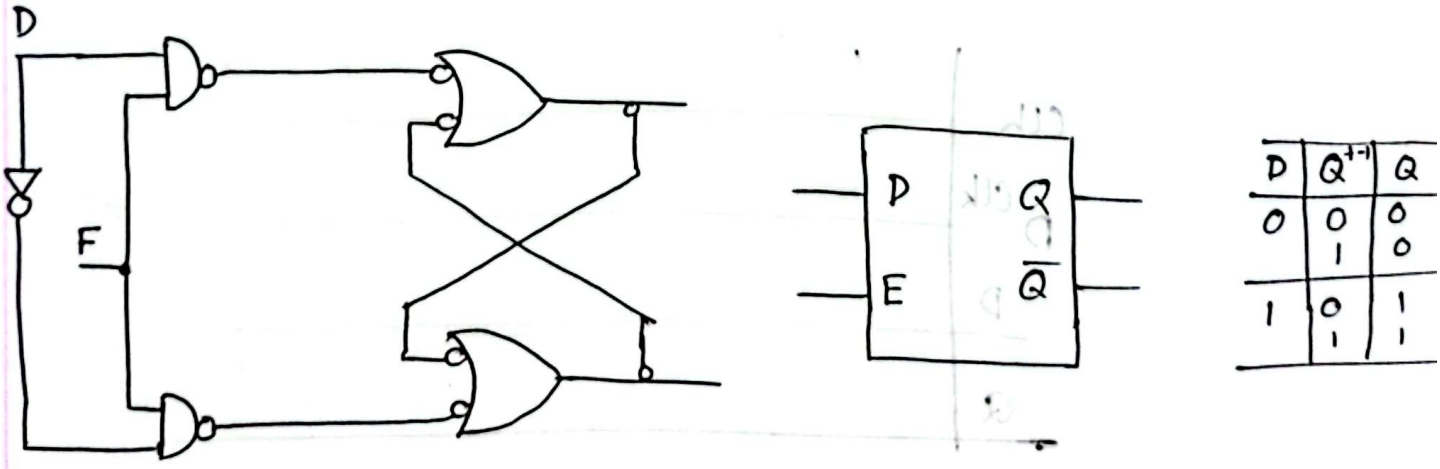
Active low





Gated S-R Latch



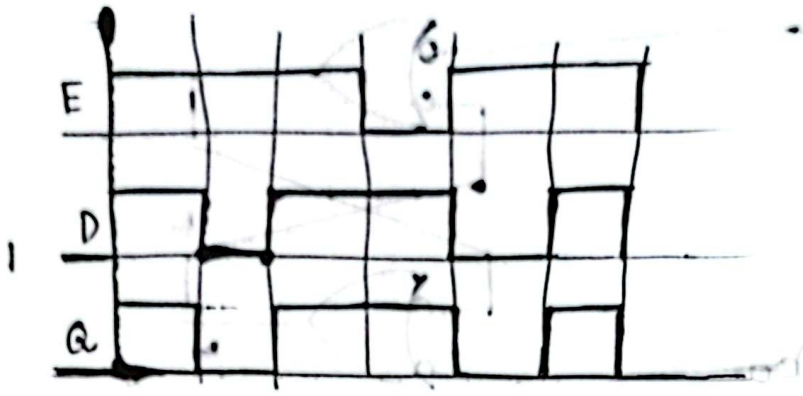


D	Q^{t+1}	Q
0	0	0
	1	0
1	0	1
	1	1

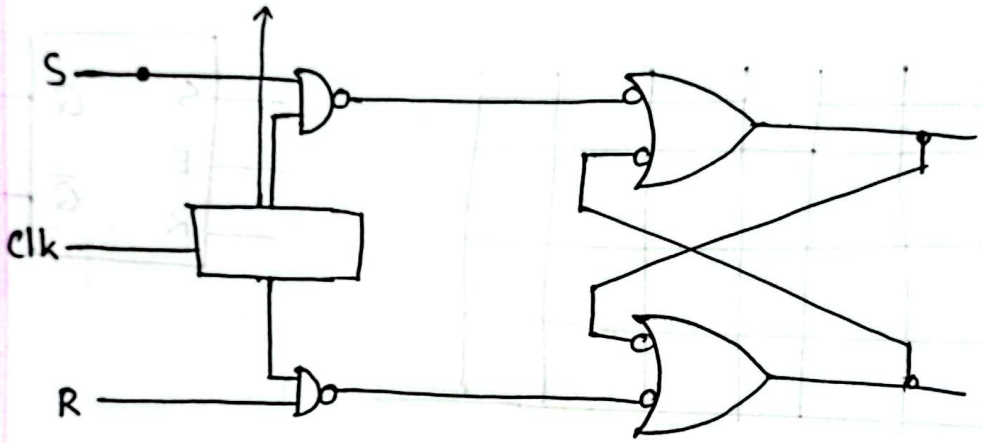
D-Latch (S-R Latch to D-latch)

D = 101101

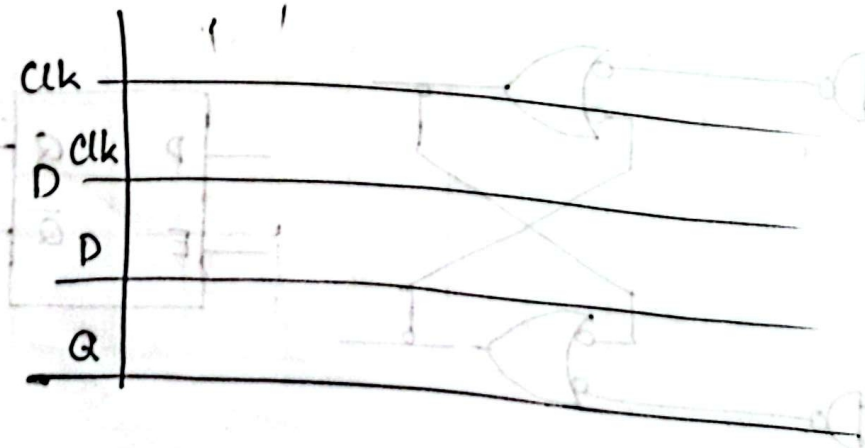
E = 111011



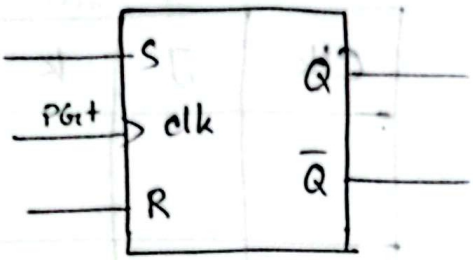
Pulse - transition ckt

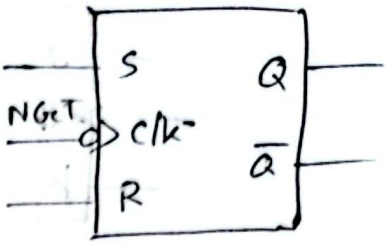


Pulse - Transition ckt

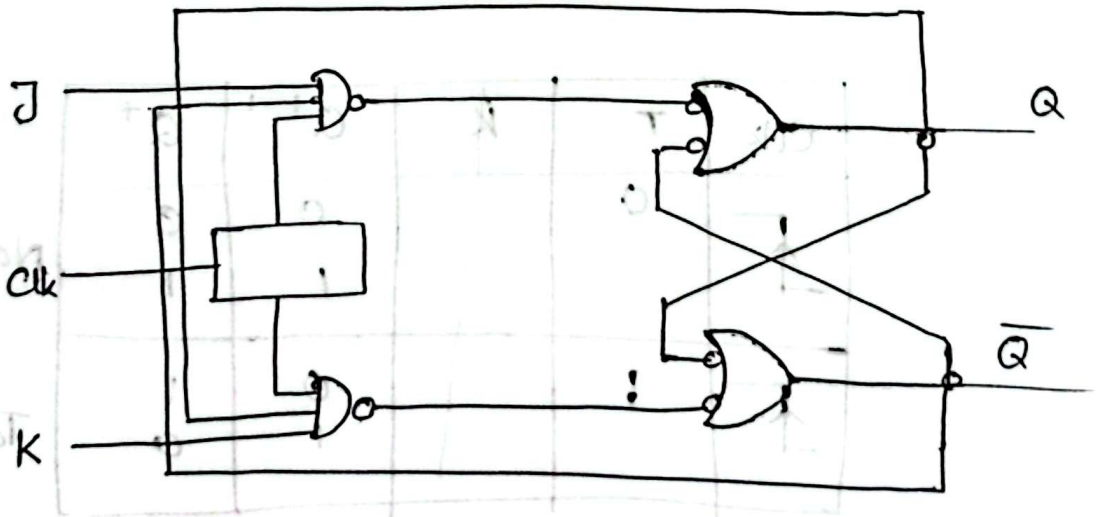


Edge-triggered Flip-Flop

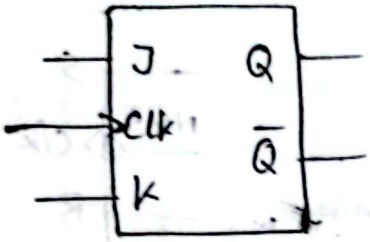




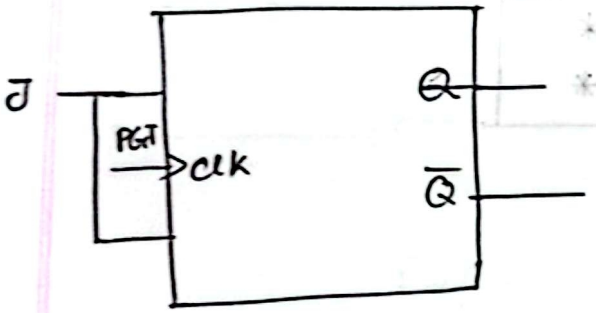
Clk	RS	CLR	Q^{+-1}	Q^{+}	
⊓	0	0	0 1	0 1	No change
⊓	1	0 1	0 1	0 0	Reset
⊓	0	0 1	1 1	1(Set) 1	Set
	1	1	0 1	* *	



J-K



Clk	J	K	Q^{t-1}	Q^t	
			0	0	No chng
			1	1	
			0	0	Reset
			1	0	
			1	1	Set
			1	1	
			0	0	Toggle
			1	1	

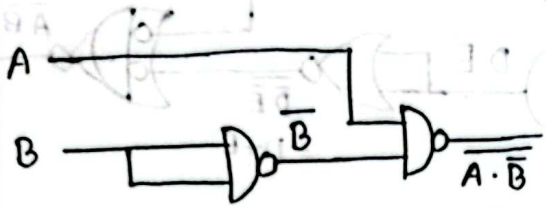


Clk	T	K	Q^{t-1}	Q^t	
↑	0		0	0	No change
			1	1	
↑	1		0	1	Toggle
			1	0	

(Master-slave Flip-Flop → Self study)

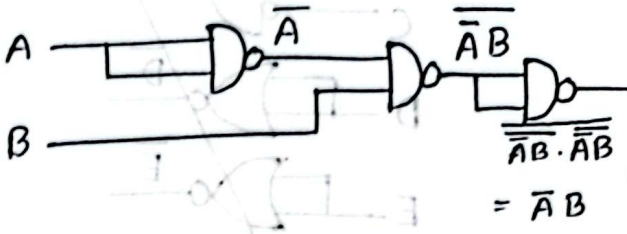
1. Use NAND to implement:

i) $X = \bar{A} + B$



$= \bar{A} + B$

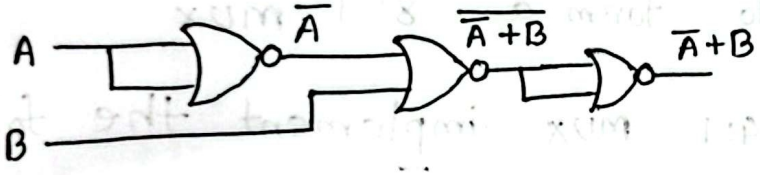
ii) $X = A\bar{B}$



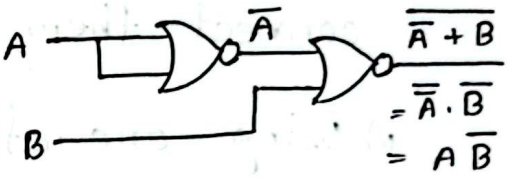
$= \bar{A}B$

2. Use NOR Gate to implement:

i) $X = \bar{A} + B$

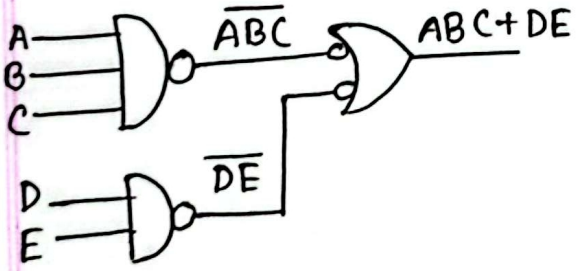


ii) $X = A\bar{B}$

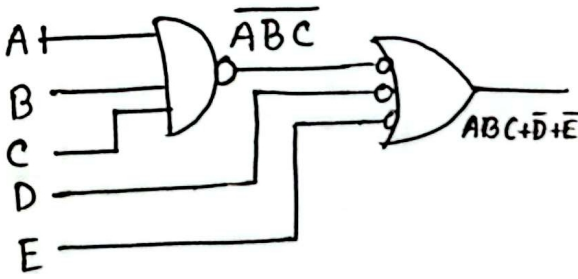


3. Implement using NAND Logic using dual symbol:

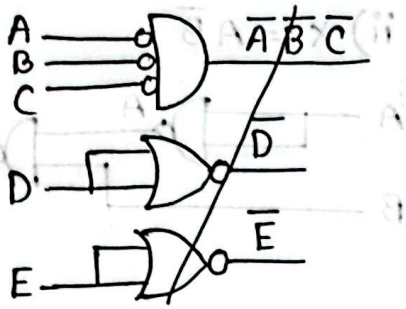
i) $ABC + DE$

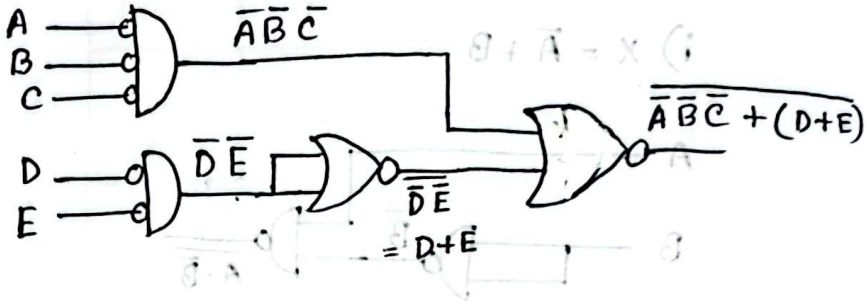


ii) $ABC + \bar{D} + \bar{E}$



4. Implement using NOR Logic : $X = \overline{\bar{A}\,\bar{B}\,\bar{C} + (D+E)}$





Lab-6 :

i) Implement two 4:1 MUX separately and ~~count~~ connect them to form a 8:1 MUX

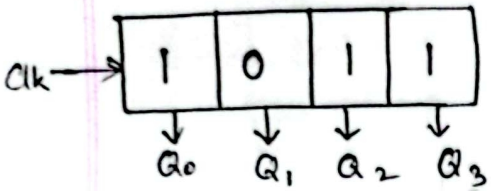
ii) Using one of 4:1 MUX implement the following boolean expression :

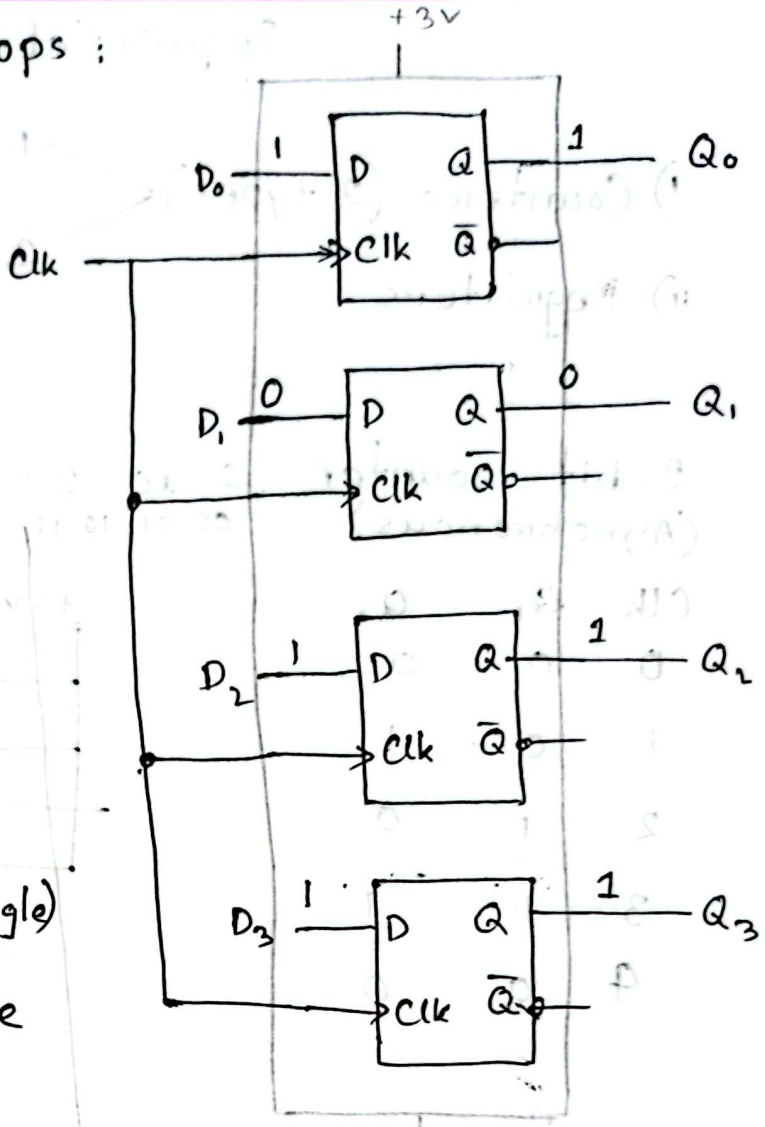
$Y = \bar{A}\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C} + ABC$

Application of Flip-flops :

D - Flip-flops :

4-bit register

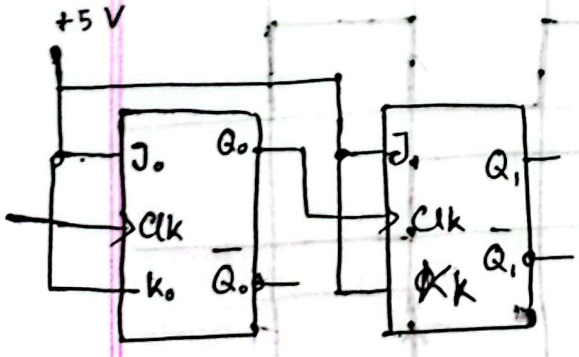


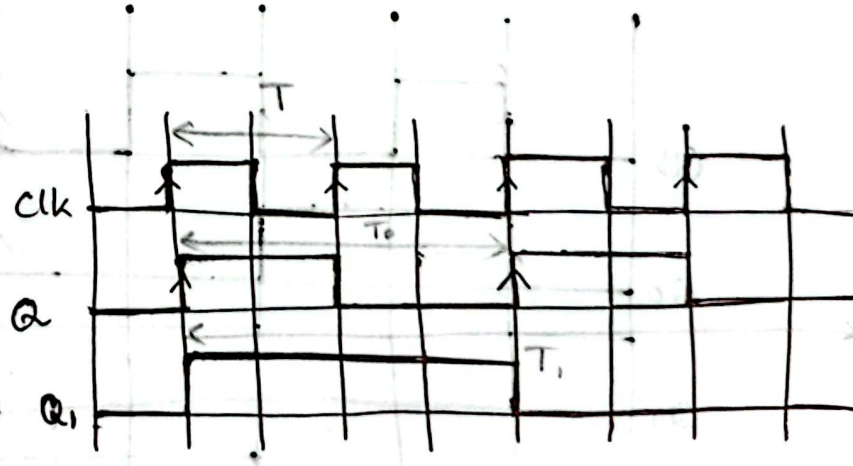


J-K Flip-Flop : (toggle)

→ frequency-ক divide করে দেয়।

Frequency divider





$f_1 = \frac{1}{T_1} = \frac{1}{2T_0} = \frac{1}{4T} = \frac{1}{4}f$

$T_1 = 2T_0 = 4T$.

$f = \frac{1}{T}$; $f_0 = \frac{1}{T_0} = \frac{1}{2T} = \frac{1}{2}f$

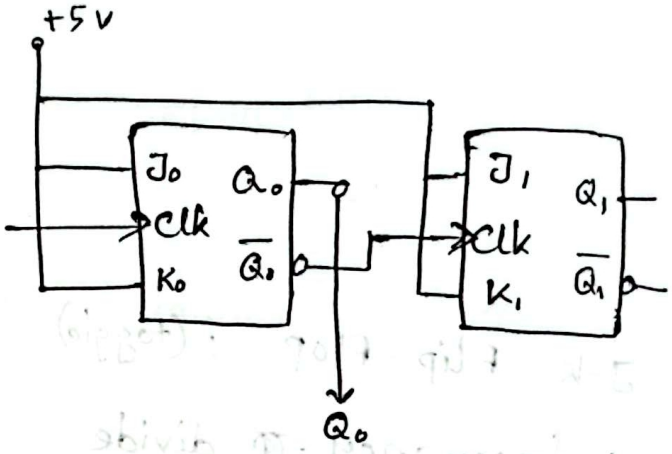
Sequential Circuits

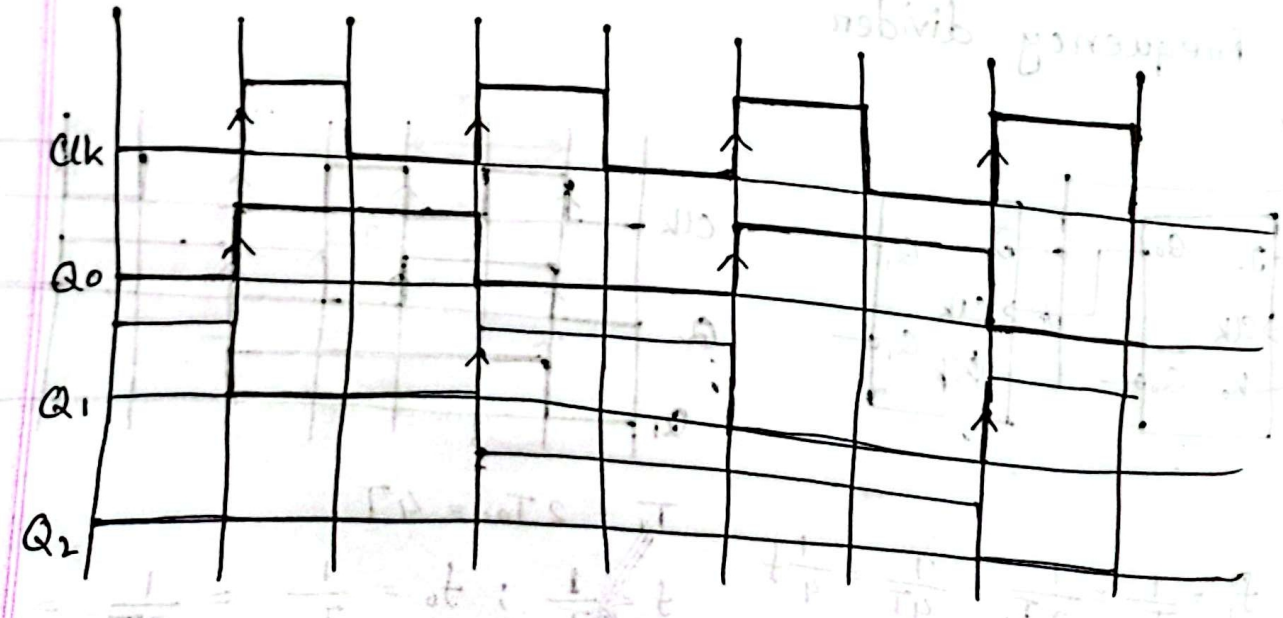
i) Counters (2 types) < 1. Asynchronous Counters
                       2. Synchronous Counters
                       (একই clock দ্বারা সবগুলো trigger হয়)

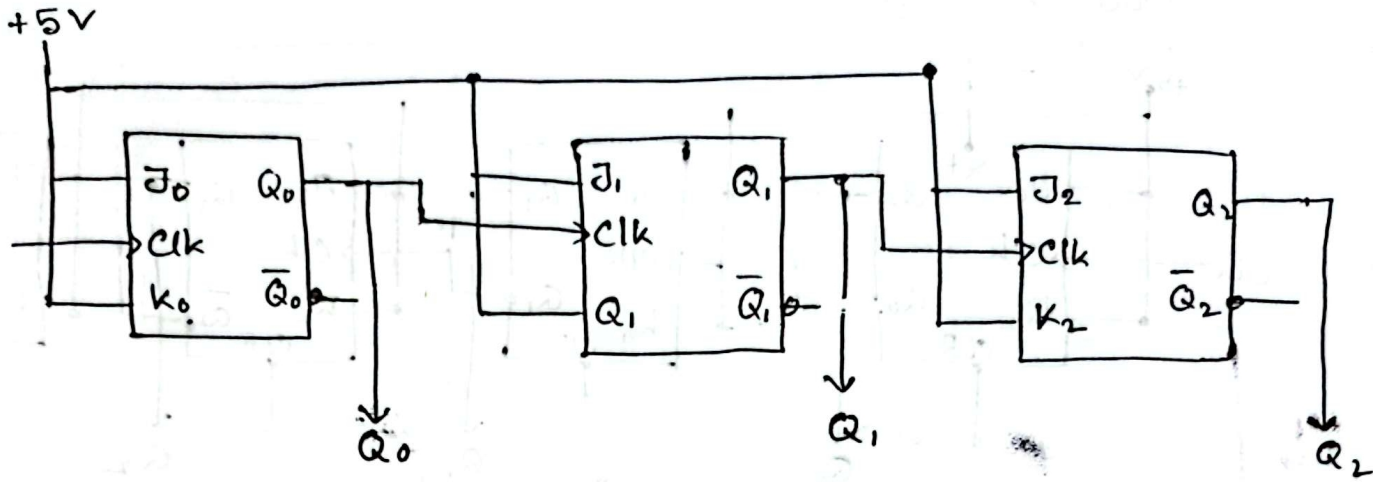
ii) Registers

2-bit counter [0 to 3 ; 4 state]
(Asynchronous) 00, 01, 10, 11

Clk	Q_1	Q_0
0	0	0
1	0	1
2	1	0
3	1	1
4	0	0

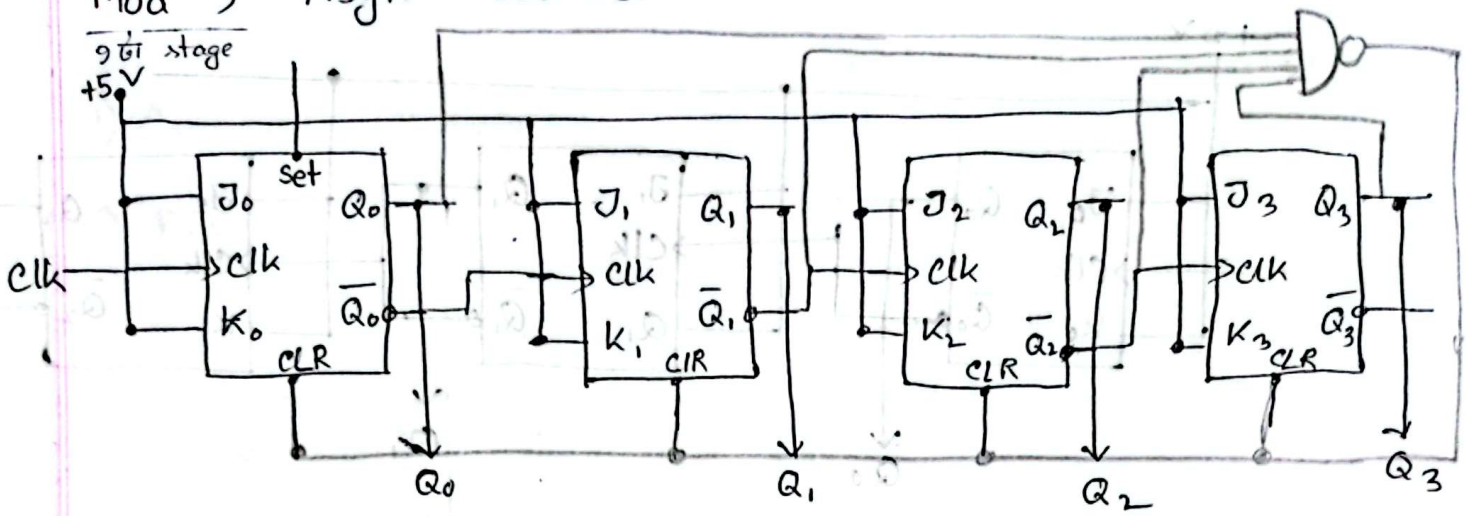






Clk	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

Mod 9 Asyn. Counter

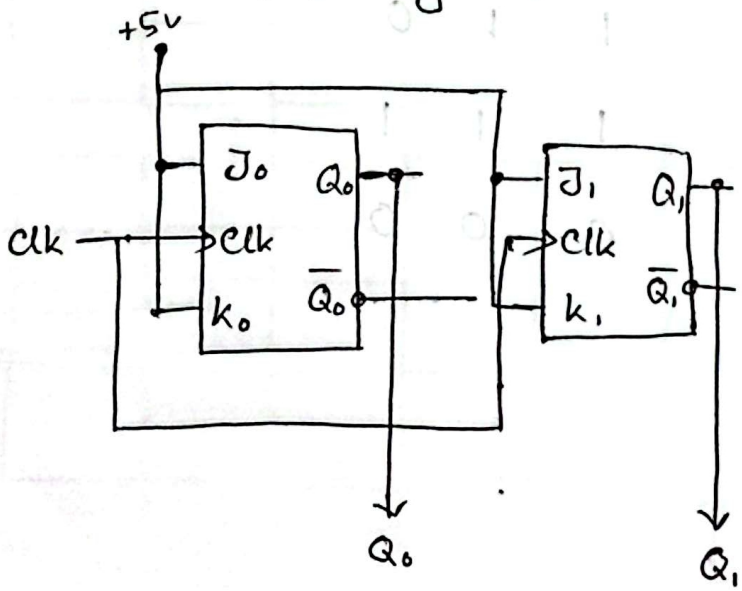


Clk	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
⋮				
8	1	0	0	0
9	1	0	0	1
15	1	1	1	1
16	0	0	0	0

Synchronous Counter

Clk	Q_1	Q_0
0	0	0
1	0	1
2	1	0
3	1	1
4	0	0

2-bit syn. Counter



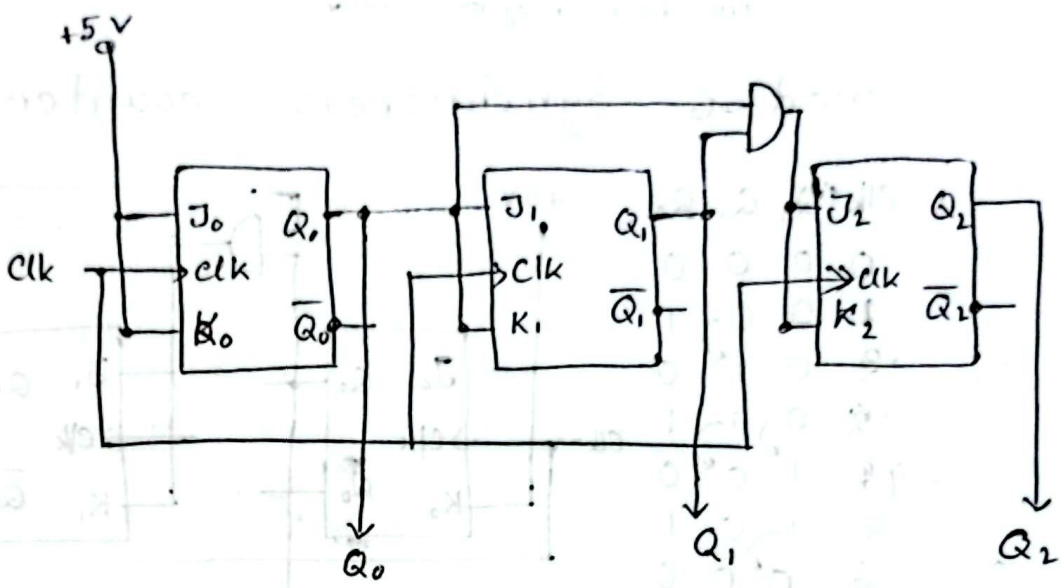
3 - bit syn. Counter (Mod 8)

$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

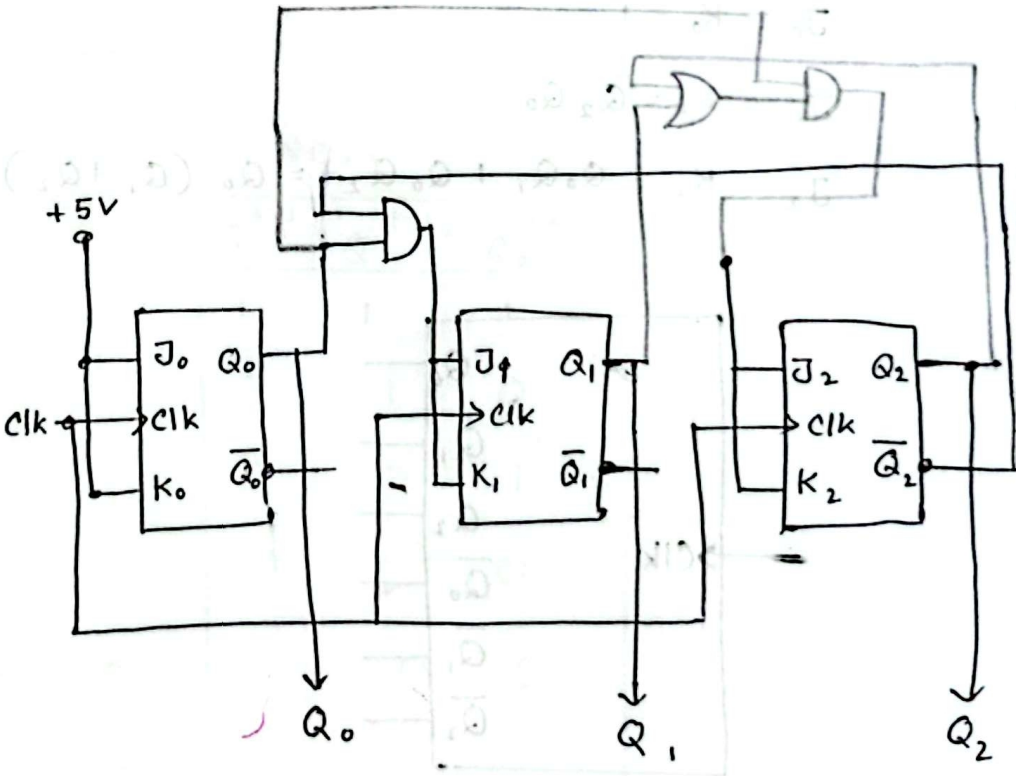
$J_2 = K_2 = Q_1 Q_0$

Clk	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0



Mod 6

Clk	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	0	0	0



$J_0 = K_0 = 1$

$J_1 = K_1 = \bar{Q}_2 Q_0$

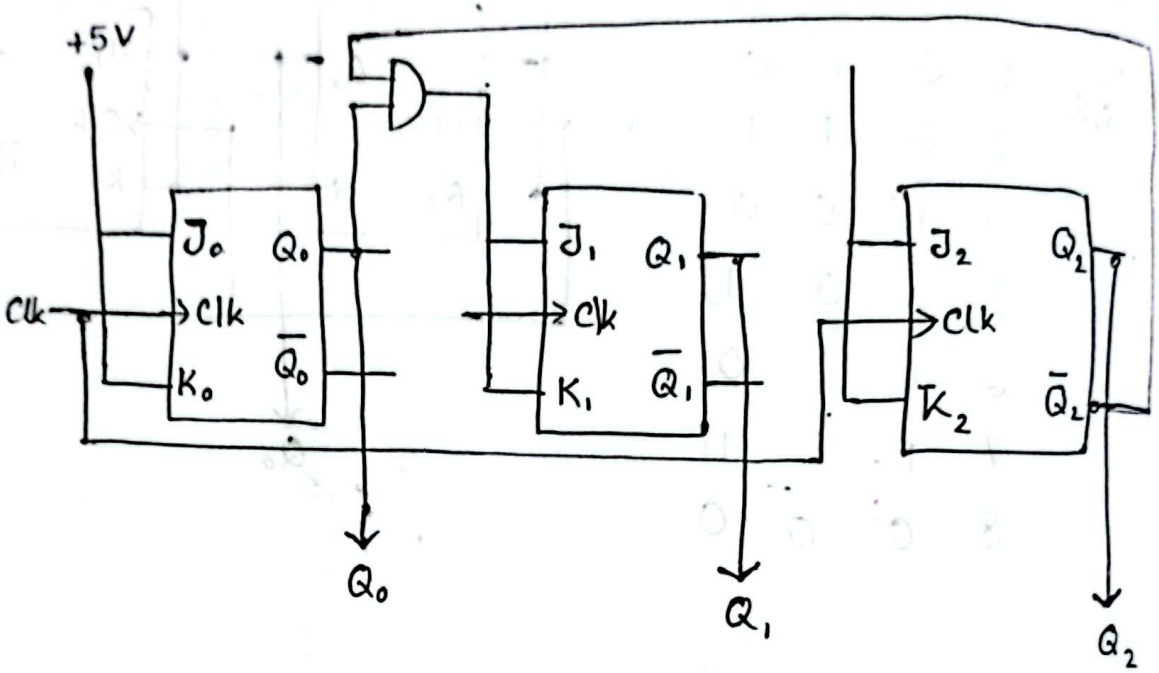
$J_2 = K_2 = Q_1 Q_0 + Q_2 Q_0$

$= Q_0 (Q_1 + Q_2)$

HW : Up down Counter, Synchronous decode Counter
Up down Irregular counter

Mod 6 Synchronous counter:

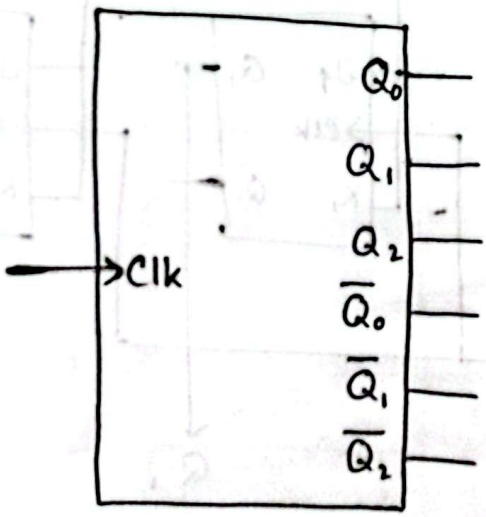
Clk	Q_2	Q_1	Q_0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	0	0	0



$J_0 = K_0 = 1$

$J_1 = K_1 = \bar{Q}_2 Q_0$

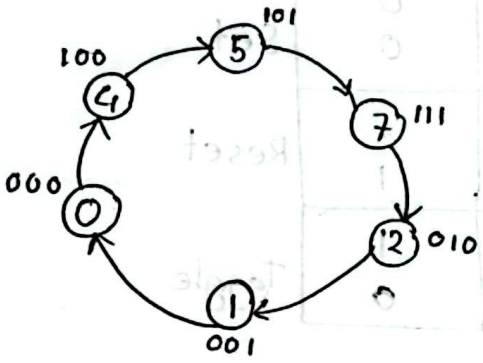
$J_2 = K_2 = Q_0 Q_1 + Q_0 Q_2 = Q_0 (Q_1 + Q_2)$



Design of Synchronous Design (Irregular)

5 - 3 - 7 - 2 - 0 - 1 - 0 - 4

1. State Diagram : (5 - 7 - 2 - 1 - 0 - 4)



2. Next State Table :

State	Present State			Next State		
	Q_2^t	Q_1^t	Q_0^t	Q_2^{t+1}	Q_1^{t+1}	Q_0^{t+1}
5	1	0	1	1	1	1
7	1	1	1	0	1	0
2	0	1	0	0	0	1
1	0	0	1	0	0	0
0	0	0	0	1	0	0
4	1	0	0	1	0	1

CamScanner

3. Truth Table of J-K flip-flop

Clk	J	K	Q^t	Q^{t+1}	
↑	0	0	0	0	No change
			1	1	
↑	0	1	0	0	Set
			1	0	
↑	1	0	0	1	Reset
			1	1	
↑	1	1	0	1	Toggle
			1	0	

(J-K flip-flop Transition st table)

Q^t	Q^{t+1}	J	K
0	0	0	*
0	1	1	*
1	0	*	1
1	1	*	0

4. Karnaugh Map

J_0

Q_2Q_1 \ Q_0	0	1
00	0	*
01	1	*
11	*	*
10	1	*

$J_0 = Q_1 + Q_2$

K_0

Q_2Q_1 \ Q_0	0	1
00	*	1
01	*	*
11	*	1
10	*	0

$K_0 = \bar{Q}_2 + Q_1$

* don't care condition

J_1 =

Q_2Q_1 \ Q_0	0	1
00	0	0
01	*	*
11	*	*
10	0	1

$J_1 = Q_2 \cdot Q_0$

K_1

Q_2Q_1 \ Q_0	0	1
00	*	*
01	1	*
11	*	0
10	*	*

$K_1 = \bar{Q}_2$

J_2

Q_2Q_1 \ Q_0	0	1
00	1	0
01	0	*
11	*	*
10	*	*

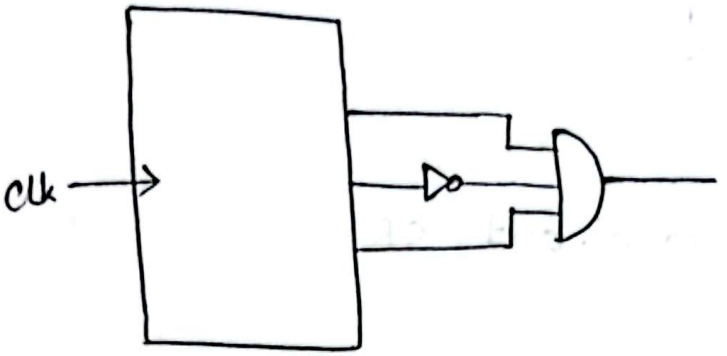
$J_2 = \bar{Q}_0 \cdot \bar{Q}_1$

K_2

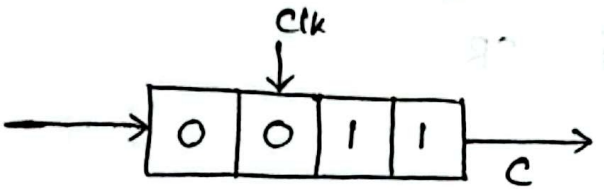
Q_2Q_1 \ Q_0	0	1
00	*	*
01	*	*
11	*	1
10	0	0

$K_2 = Q_1$

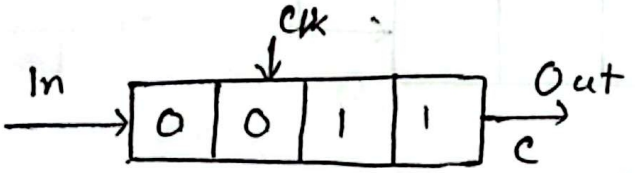
Decoding by Counter Decoding



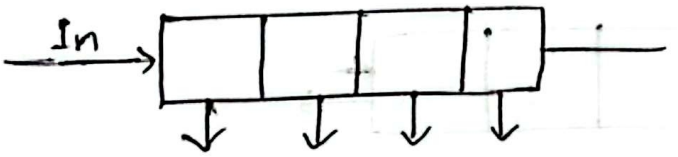
Shift Registors ← storing device



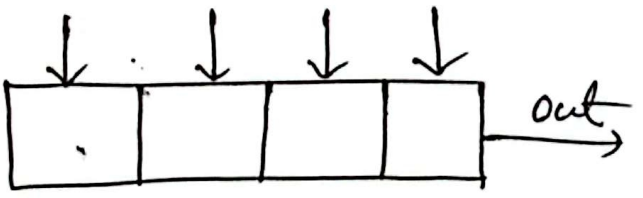
right shift registor



1. Serial-in-serial-out shift registor

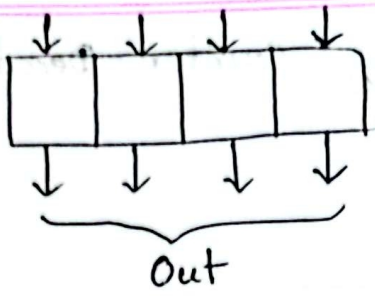


2. Serial-in-parallel out shift registor

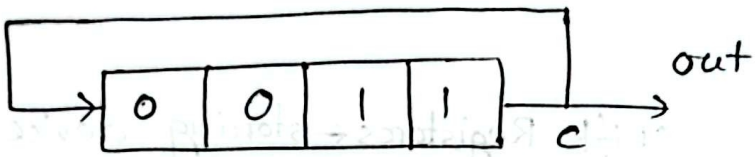


3. Parallel-in-serial out SR

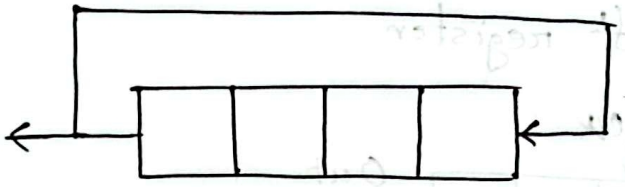
CamScanner



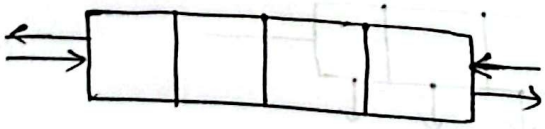
4. Parallel - in- parallel-out SR



5. Rotate - right SR

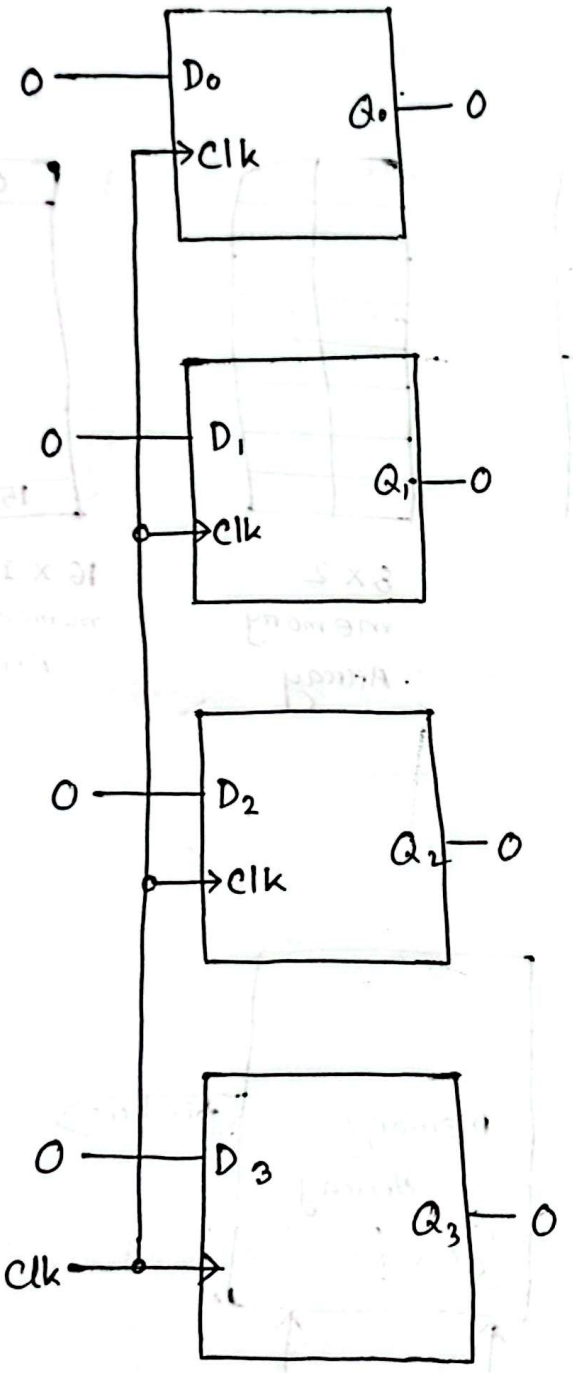


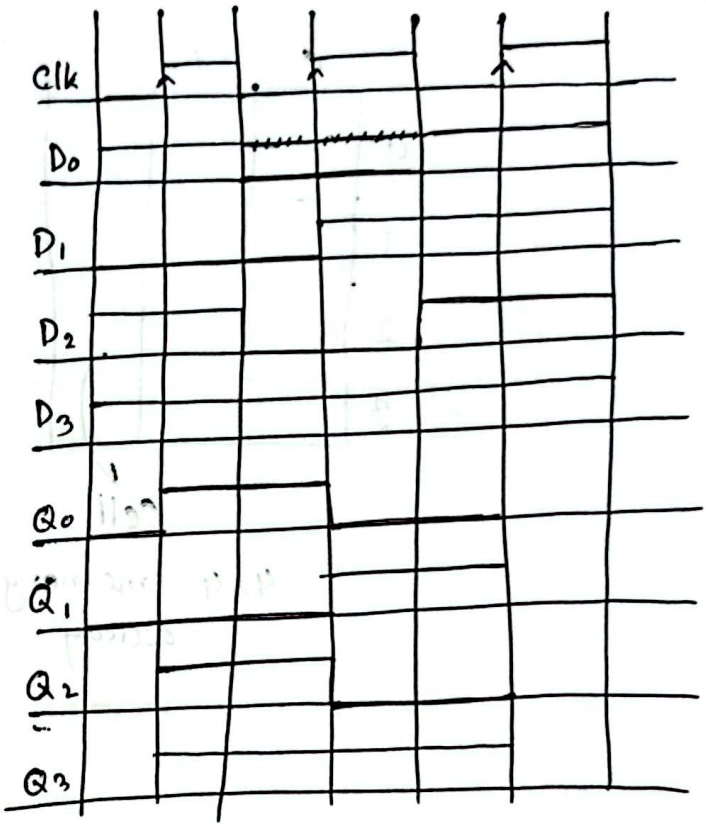
6. Rotate - left. SR



7. Bi-directional SR

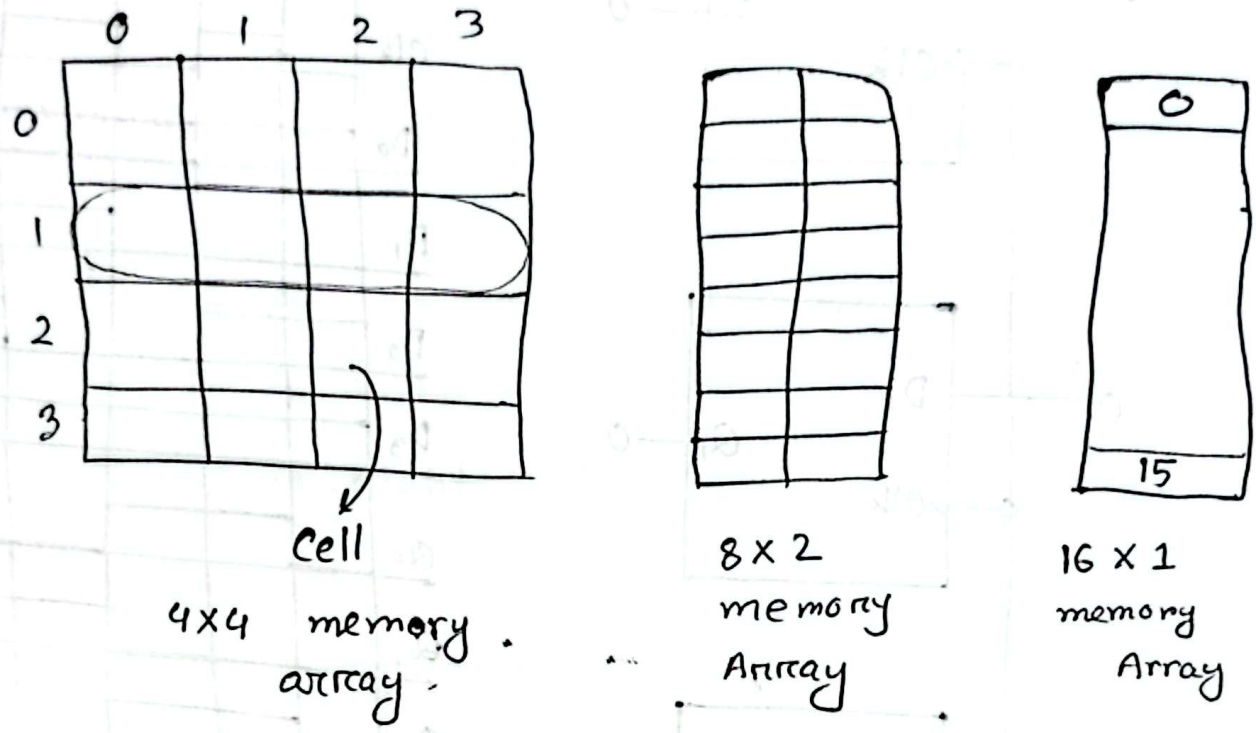
4-bit Parallel-in-Parallel-out SR

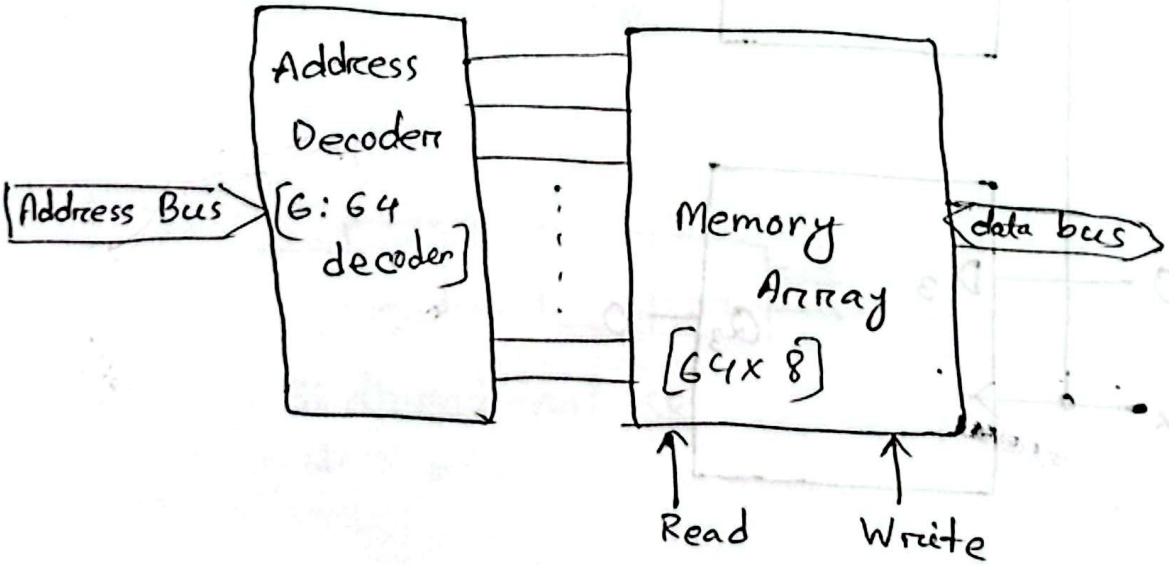




Data Storage

Bit, nibble, byte, word





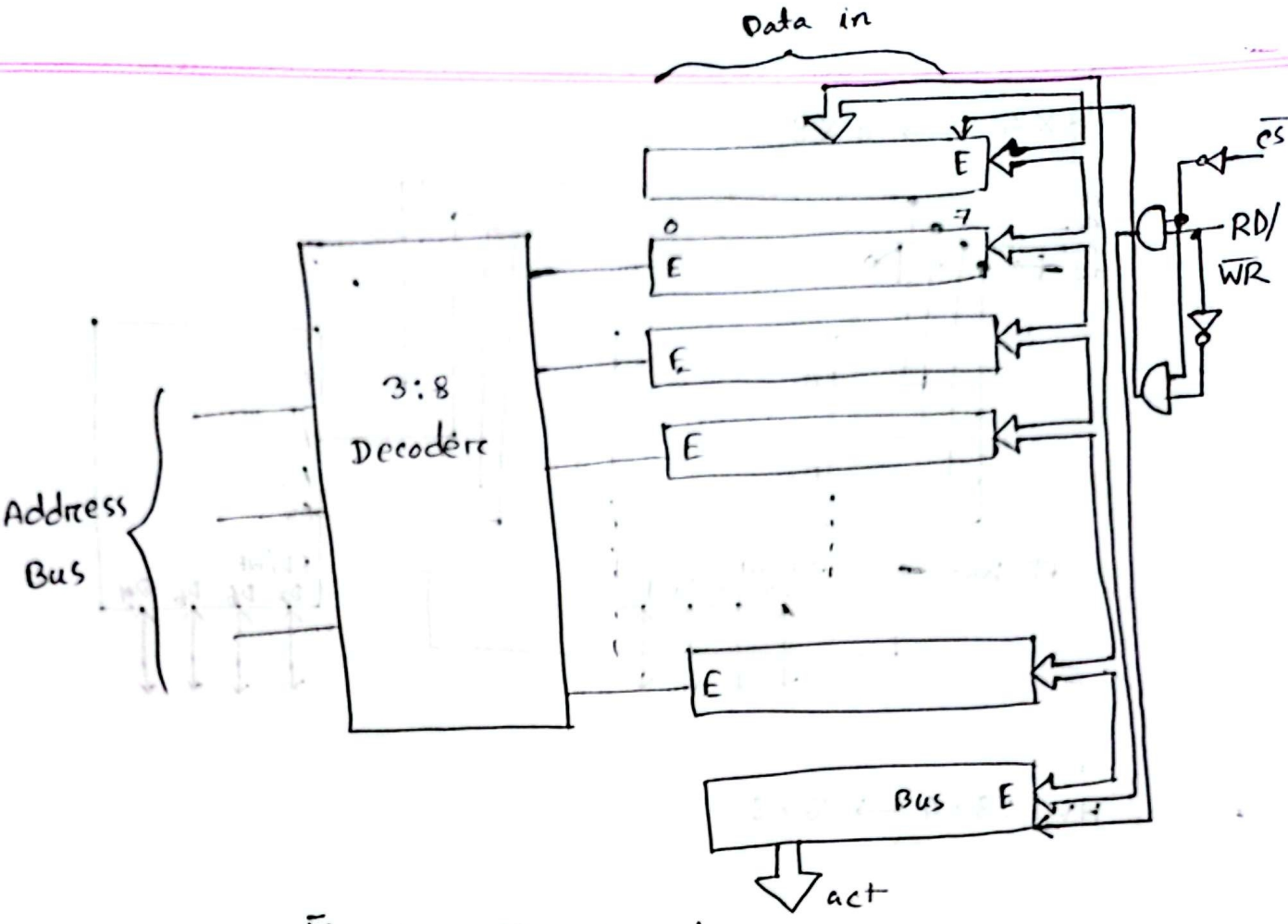
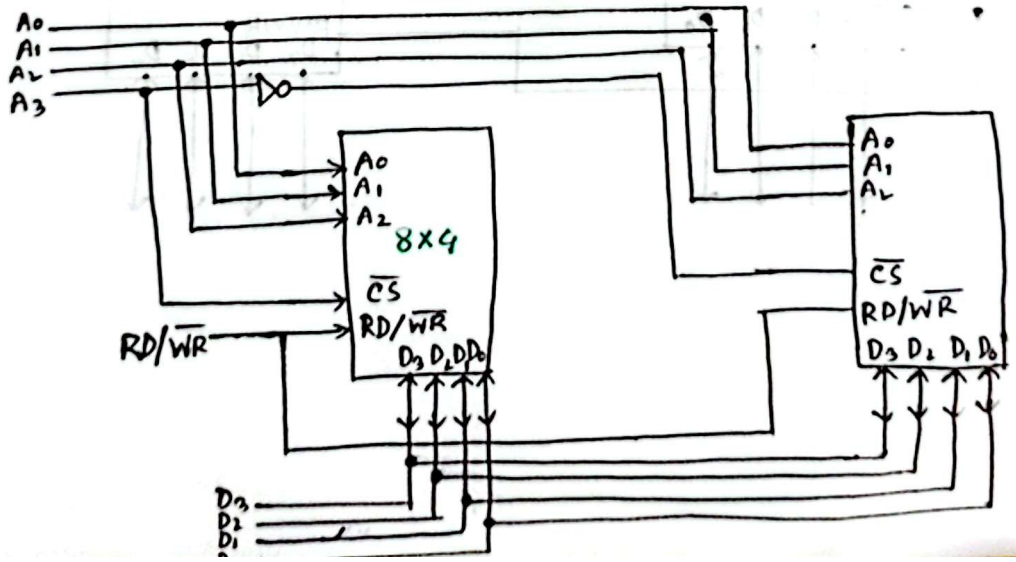


Fig : 8×4 Memory Array

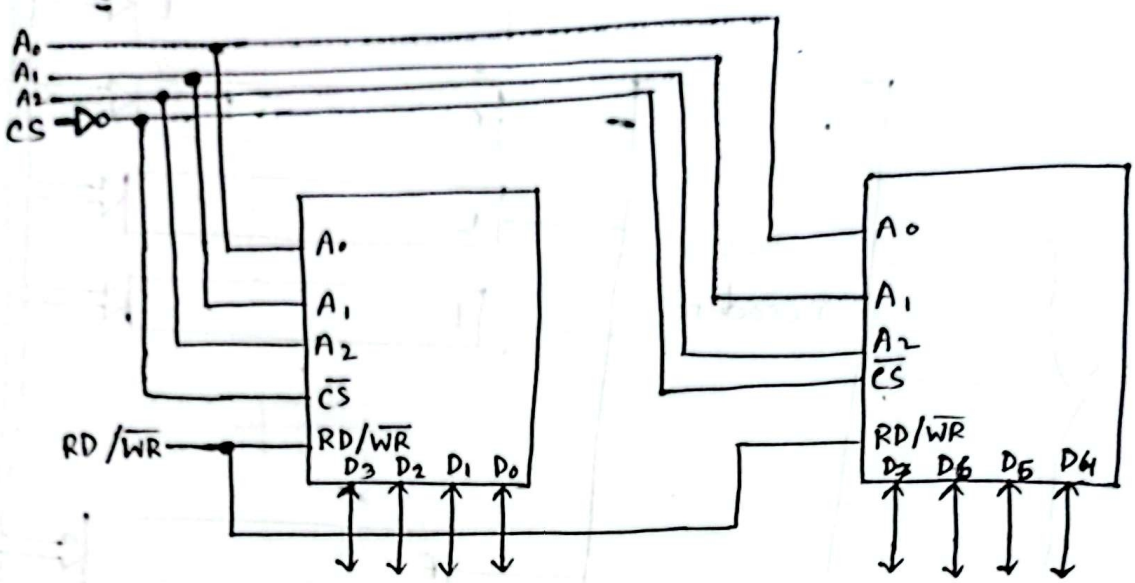
Expansion of Memory Module :

$8 \times 4 \longrightarrow 16 \times 4$

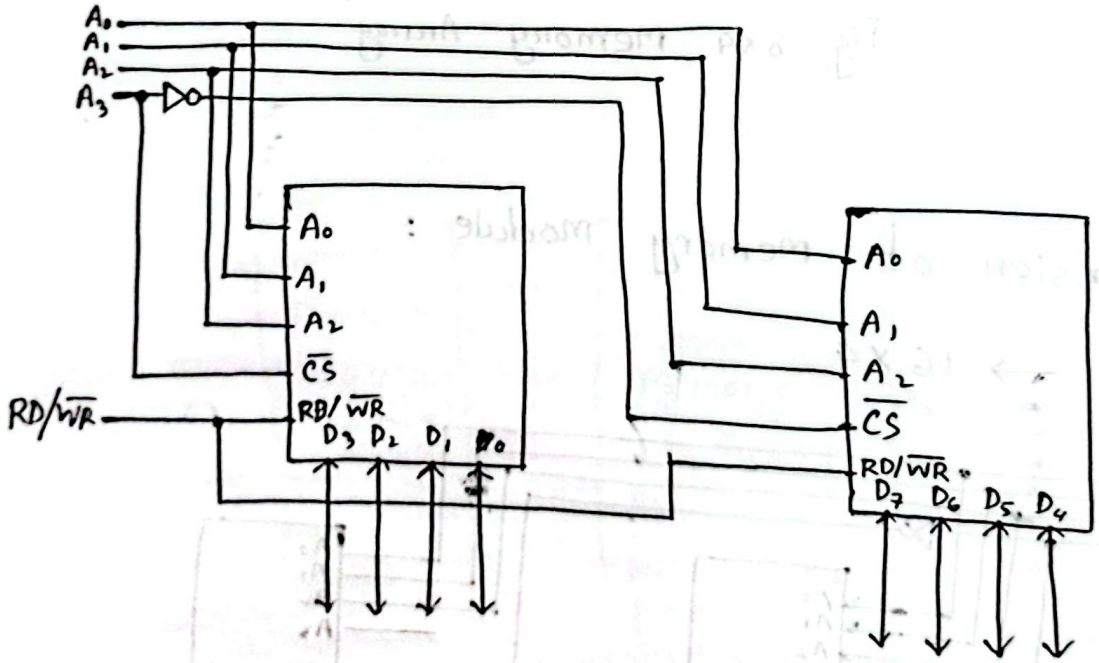
2^3 2^4



8 × 4 ⟶ 8 × 8

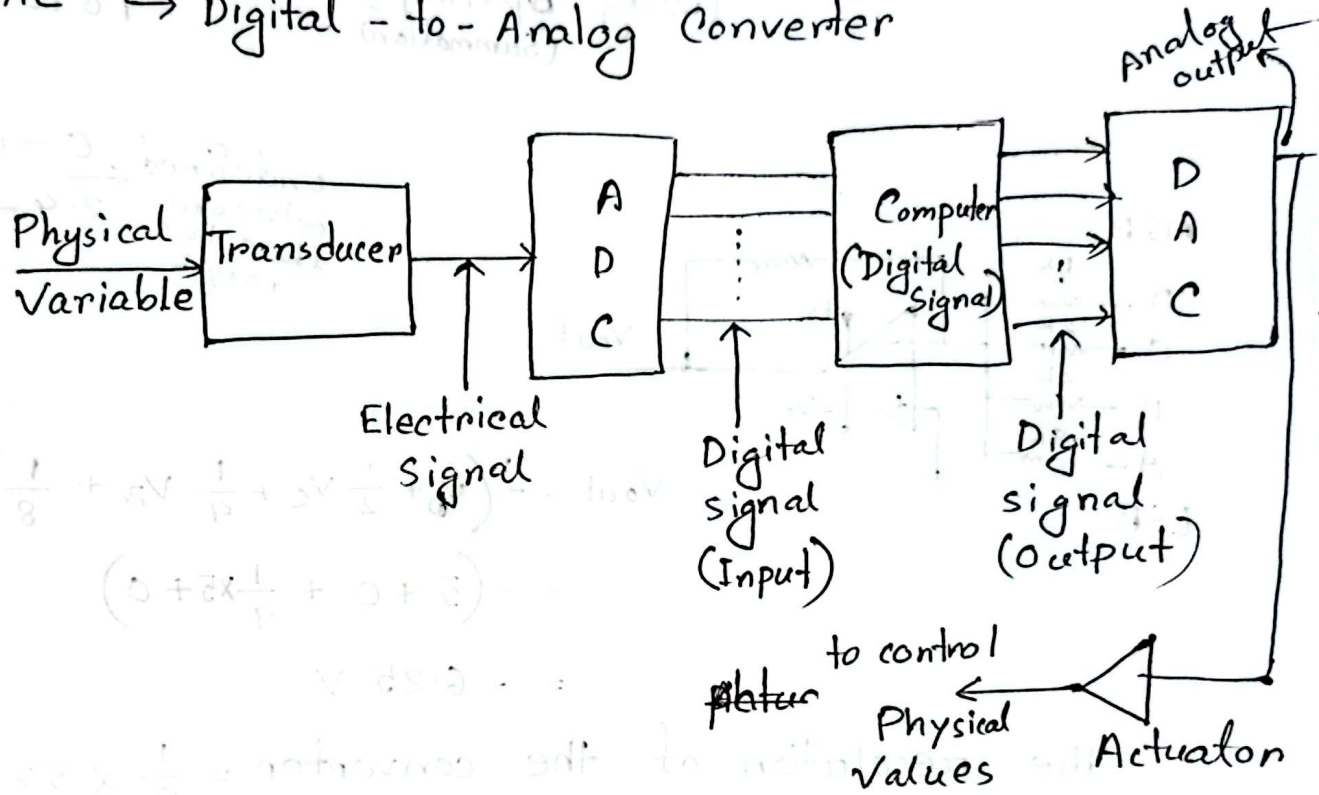


HW : 8 × 4 ⟶ 16 × 8



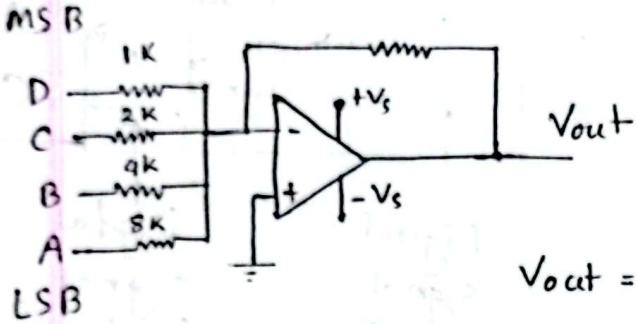
ADC → Analog - to - Digital Converter

DAC → Digital - to - Analog Converter



Simple DAC
(Using Op-Amp)
(Summation)

DCBA
1010



Undefined between these two range ← 0 − 0.4 → 0 V
2.4 − 5 → 1 V

$$V_{out} = -\left(V_D + \frac{1}{2}V_C + \frac{1}{4}V_B + \frac{1}{8}V_A\right)$$

$$= -\left(5 + 0 + \frac{1}{4}\times 5 + 0\right)$$

$$= -6.25\ V$$

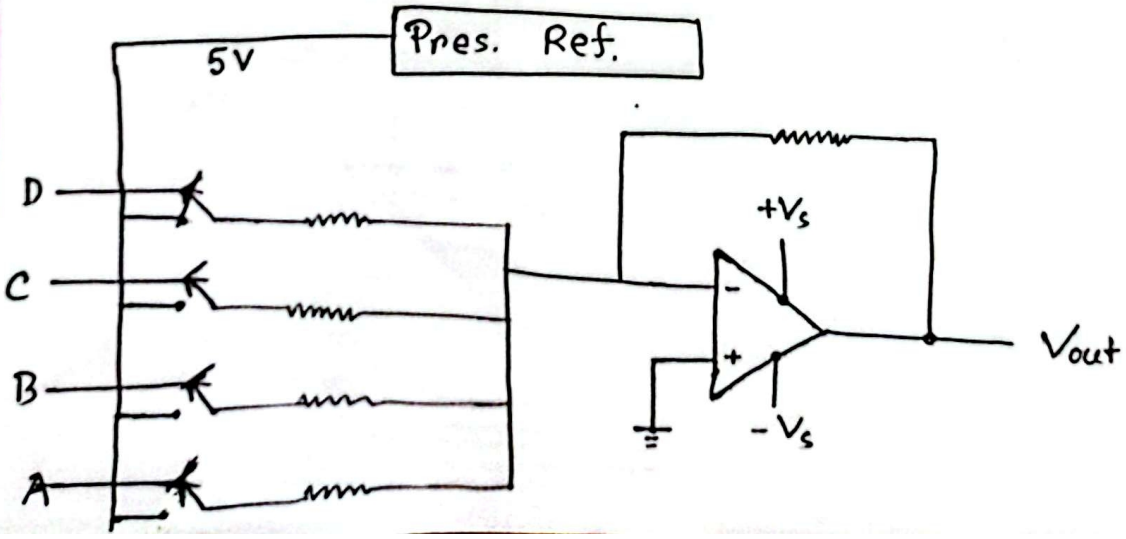
The resolution of the converter $= \frac{1}{8} \times 5V$

$= 0.625\ V$

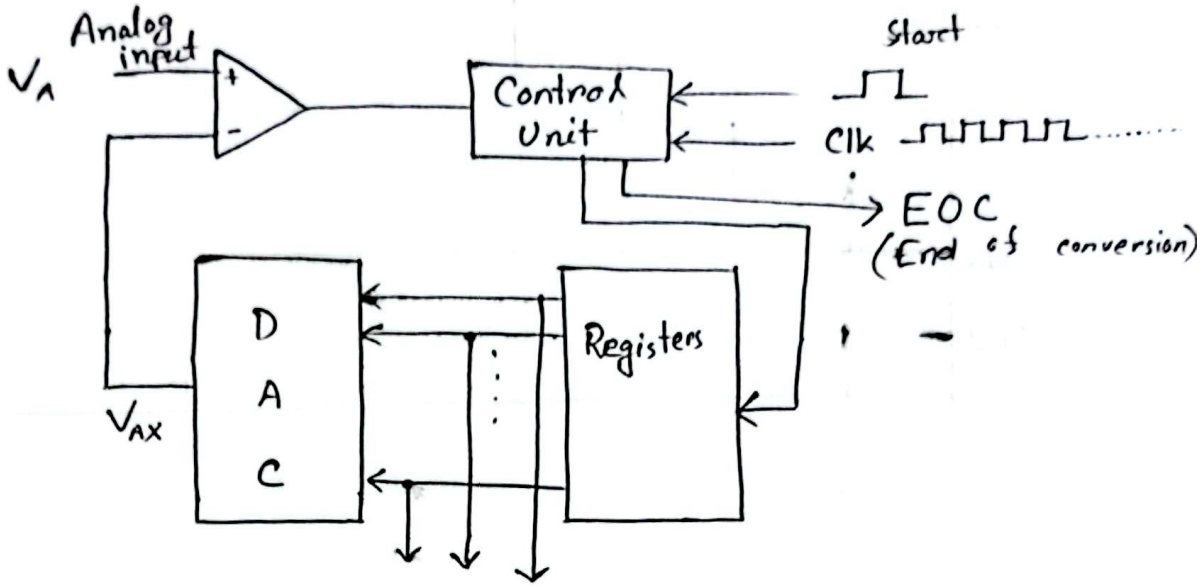
$$\text{Analog Output} = \frac{V_{out}}{\text{resolution}} = \frac{-6.25}{0.625} = -10$$

Precession Voltage

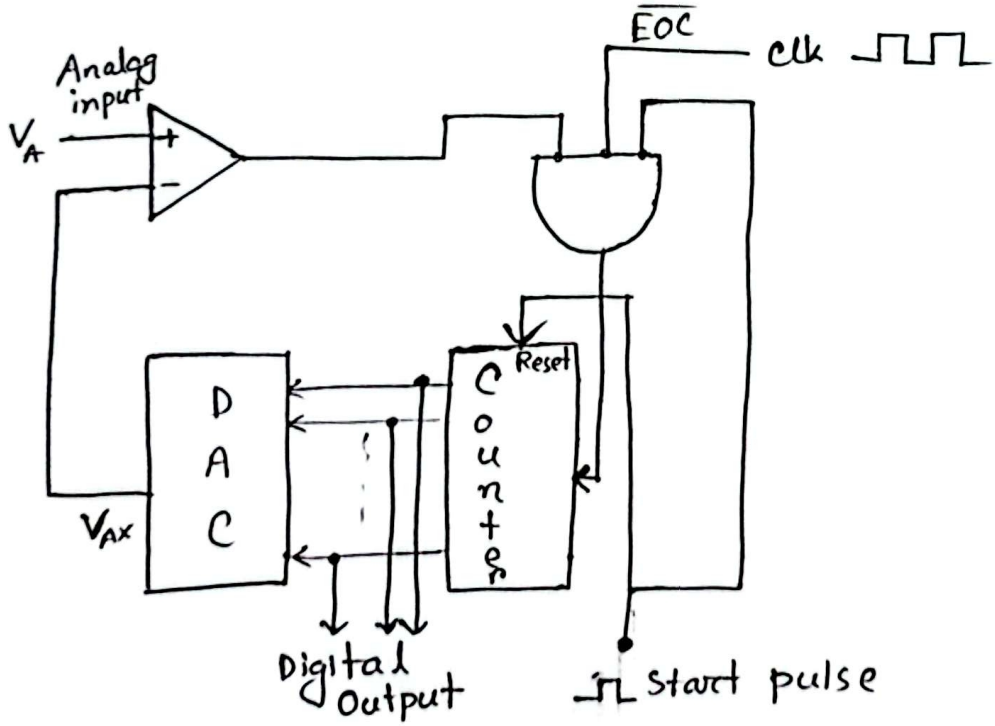
DAC with precession reference Supply

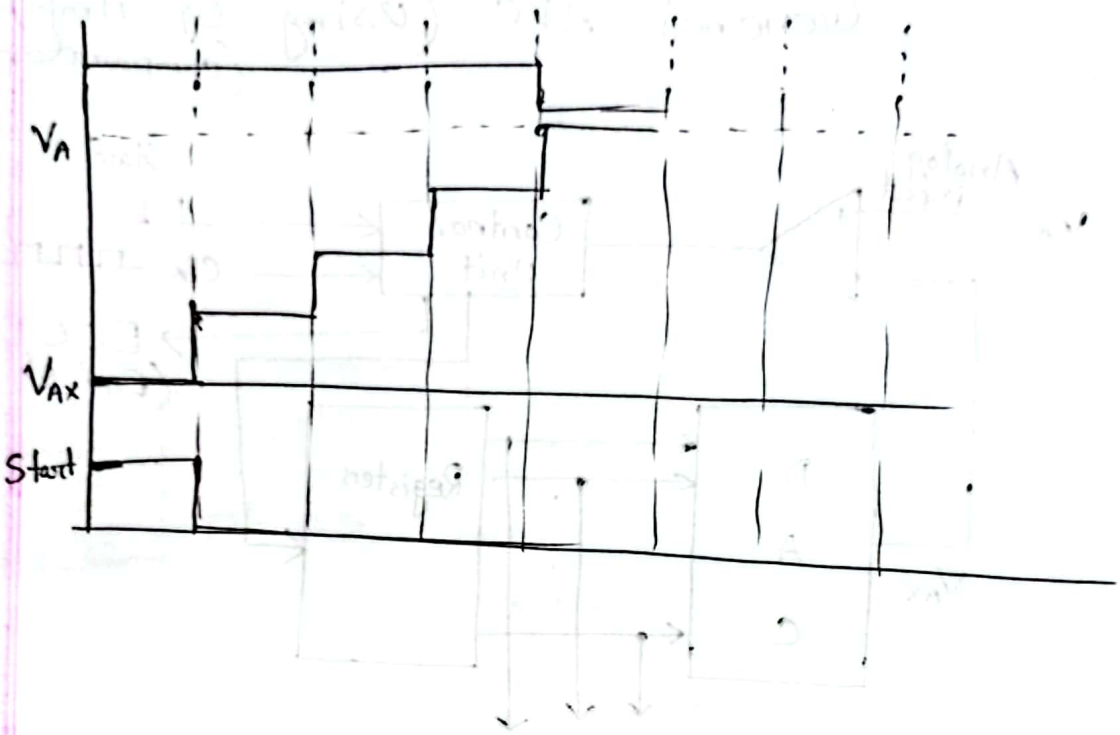


General ADC (Using Op-Amp (comparator))



Digital Ramp ADC




V_A
V_{AX}
Start